Complex Analysis I

Lecture notes following Stein–Shakarchi, Ch. 1–7 (through the prime number theorem)

Contents

Chapter 1. Preliminaries to Complex Analysis 2

1 Complex numbers and the complex plane 2

1.1 Basic properties . . . 2

1.2 Convergence . . . 2

1.3 Sets in the complex plane . . . 3

2 Functions on the complex plane 4

2.1 Continuous functions . . . 4

2.2 Holomorphic functions . . . 4

2.3 Power series . . . 6

3 Integration along curves 8

Chapter 2. Cauchy's Theorem and Its Applications 10

1 Goursat's theorem 10

2 Local existence of primitives and Cauchy's theorem in a disc 11

3 Evaluation of some integrals 13

4 Cauchy's integral formulas 13

5 Further applications 17

5.1 Morera's theorem . . . 17

5.2 Sequences of holomorphic functions . . . 17

5.3 Holomorphic functions defined in terms of integrals . . . 18

5.4 Schwarz reflection principle . . . 18

5.5 Runge's approximation theorem . . . 19

Chapter 3. Meromorphic Functions and the Logarithm 21

1 Zeros and poles 21

2 The residue formula **22**
2.1 Examples . . . 23

3 Singularities and meromorphic functions **25**

4 The argument principle and applications **26**

5 Homotopies and simply connected domains **28**

6 The complex logarithm **29**

7 Fourier series and harmonic functions **30**

Chapter 4. The Fourier Transform **32**

1 The class $\mathfrak{F}$ **32**

2 Action of the Fourier transform on $\mathfrak{F}$ **32**

3 Paley–Wiener theorem **34**

Chapter 5. Entire Functions **37**

1 Jensen's formula **37**

2 Functions of finite order **38**

3 Infinite products **39**
3.1 Generalities . . . 39
3.2 Example: the product formula for the sine function . . . 40

4 Weierstrass infinite products **40**

5 Hadamard's factorization theorem **41**

Chapter 6. The Gamma and Zeta Functions **44**

1 The gamma function **44**
1.1 Analytic continuation . . . 44
1.2 Further properties of Γ . . . 45

2 The zeta function **47**
2.1 Functional equation and analytic continuation . . . 47

Chapter 7. The Zeta Function and Prime Number Theorem **51**

1 Zeros of the zeta function **51**

1.1 Estimates for $1/\zeta(s)$ 53

2 Reduction to the functions ψ and ψ_1 **53**

2.1 Proof of the asymptotics for ψ_1 55

Note on interchanging double sums **55**

Chapter 1. Preliminaries to Complex Analysis

Overview

We review the algebraic and analytic properties of complex numbers and some topology of the complex plane. We then define *holomorphicity*, the complex analytic version of differentiability, which leads to the Cauchy–Riemann equations and to power series. Finally we define curves and integrals along them, and prove a first result in the direction of Cauchy's theorem: if f has a primitive (a holomorphic F with $F' = f$), then $\int_\gamma f(z)\,dz = 0$ for every closed curve γ.

1 Complex numbers and the complex plane

1.1 Basic properties

A complex number has the form $z = x+iy$ with x, y real and $i^2 = -1$; $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$ are its **real part** and **imaginary part**. Real numbers are those with zero imaginary part; a number with zero real part is **purely imaginary**. The set of complex numbers is $\mathbb{C}$, identified with $\mathbb{R}^2$ by $z = x + iy \leftrightarrow (x, y)$; the x- and y-axes are the **real axis** and **imaginary axis**.

Treating all numbers as real and using $i^2 = -1$: if $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$,

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2), \qquad z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2).$$

These operations are commutative, associative and distributive. Addition is vector addition in $\mathbb{R}^2$; multiplication is a rotation composed with a dilation (see the polar form below); multiplication by i is rotation by $\pi/2$.

The **absolute value** of $z = x+iy$ is $|z| = (x^2+y^2)^{1/2}$, the euclidean distance from 0 to (x, y). The triangle inequality $|z + w| \le |z| + |w|$ holds, and also $|\mathrm{Re}(z)| \le |z|$, $|\mathrm{Im}(z)| \le |z|$ and

$$\big||z| - |w|\big| \le |z - w|,$$

which follows from $|z| \le |z - w| + |w|$ and $|w| \le |z - w| + |z|$.

The **complex conjugate** of $z = x + iy$ is $\overline{z} = x - iy$ (reflection across the real axis). z is real iff $z = \overline{z}$, purely imaginary iff $z = -\overline{z}$, and

$$\mathrm{Re}(z) = \frac{z + \overline{z}}{2}, \qquad \mathrm{Im}(z) = \frac{z - \overline{z}}{2i}, \qquad |z|^2 = z\overline{z}, \qquad \frac{1}{z} = \frac{\overline{z}}{|z|^2}\ (z \neq 0).$$

Any non-zero z has a **polar form** $z = re^{i\theta}$ with $r > 0$, $\theta \in \mathbb{R}$, where $e^{i\theta} = \cos\theta + i\sin\theta$. Since $|e^{i\theta}| = 1$, $r = |z|$; θ is the **argument** $\arg z$, defined up to a multiple of 2π, the counterclockwise angle from the positive real axis to the half-line through z. If $z = re^{i\theta}$ and $w = se^{i\varphi}$ then $zw = rse^{i(\theta+\varphi)}$: multiplication is a homothety (a rotation composed with a dilation).

1.2 Convergence

A sequence $\{z_n\}$ **converges** to $w \in \mathbb{C}$ if $\lim_{n\to\infty}|z_n - w| = 0$; we write $w = \lim z_n$. Since $|\cdot|$ on $\mathbb{C}$ is the euclidean distance on $\mathbb{R}^2$, this is convergence of the points in the plane; equivalently, the real and imaginary parts of z_n converge to those of w. $\{z_n\}$ is a **Cauchy sequence** if $|z_n - z_m| \to 0$ as $n, m \to \infty$, i.e. for every $\epsilon > 0$ there is N with $|z_n - z_m| < \epsilon$ for $n, m > N$. $\mathbb{R}$ is complete (every Cauchy sequence of reals converges), and $\{z_n\}$ is Cauchy iff its real and imaginary parts are; hence:

Theorem 1.1

$\mathbb{C}$, the complex numbers, is complete.

1.3 Sets in the complex plane

For $z_0 \in \mathbb{C}$ and $r > 0$, the **open disc** is $D_r(z_0) = \{z \in \mathbb{C} : |z - z_0| < r\}$, the **closed disc** is $\overline{D}_r(z_0) = \{z : |z - z_0| \leq r\}$, and their boundary is the circle $C_r(z_0) = \{z : |z - z_0| = r\}$. The unit disc is denoted $\mathbb{D} = \{z : |z| < 1\}$.

Given $\Omega \subset \mathbb{C}$, z_0 is an **interior point** of Ω if $D_r(z_0) \subset \Omega$ for some $r > 0$; the **interior** of Ω consists of its interior points. Ω is **open** if every point of Ω is an interior point, and **closed** if $\Omega^c = \mathbb{C} - \Omega$ is open. z is a **limit point** of Ω if there are $z_n \in \Omega$, $z_n \neq z$, with $z_n \to z$; a set is closed iff it contains all its limit points. The **closure** $\overline{\Omega}$ is Ω together with its limit points, and the **boundary** $\partial\Omega$ is the closure minus the interior. Ω is **bounded** if $|z| < M$ for $z \in \Omega$; then $\operatorname{diam}(\Omega) = \sup_{z,w\in\Omega}|z - w|$. Ω is **compact** if it is closed and bounded.

Theorem 1.2

The set $\Omega \subset \mathbb{C}$ is compact if and only if every sequence $\{z_n\} \subset \Omega$ has a subsequence that converges to a point in Ω.

Proof [Supplement]. If Ω is closed and bounded, a sequence in Ω has bounded real and imaginary parts; by Bolzano–Weierstrass (twice) a subsequence converges, and its limit lies in Ω since Ω is closed. Conversely, if Ω were unbounded, choose $z_n \in \Omega$ with $|z_n| > n$; no subsequence converges. If Ω were not closed, some limit point $z \notin \Omega$ is the limit of $z_n \in \Omega$, and every subsequence converges to $z \notin \Omega$. ■

An **open covering** of Ω is a family of open sets $\{U_\alpha\}$ with $\Omega \subset \bigcup_\alpha U_\alpha$.

Theorem 1.3

A set Ω is compact if and only if every open covering of Ω has a finite subcovering.

Proof [Supplement]. Suppose Ω is compact and $\{U_\alpha\}$ is an open covering with no finite subcovering. Ω lies in a square Q_0; subdivide it into four equal squares; the parts of Ω in at least one of them, Q_1, cannot be finitely covered. Repeating, we get nested squares Q_n with $\operatorname{diam}(Q_n) \to 0$ such that $\Omega \cap Q_n$ is not finitely covered. Pick $z_n \in \Omega \cap Q_n$; $\{z_n\}$ is Cauchy, converges to some $w \in \Omega$ (Ω is closed), and $w \in U_\alpha$ for some α. Since U_α is open and $\operatorname{diam}(Q_n) \to 0$, $Q_n \subset U_\alpha$ for large n, a contradiction. Conversely, if every open covering has a finite subcovering, the discs $D_n(0)$ cover Ω, so Ω is bounded; and if $z \notin \Omega$ were a limit point, the sets $\{w : |w - z| > 1/n\}$ would cover Ω with no finite subcovering. ■

Proposition 1.4

If $\Omega_1 \supset \Omega_2 \supset \cdots \supset \Omega_n \supset \cdots$ is a sequence of non-empty compact sets in $\mathbb{C}$ with the property that $\operatorname{diam}(\Omega_n) \to 0$ as $n \to \infty$, then there exists a unique point $w \in \mathbb{C}$ such that $w \in \Omega_n$ for all n.

Proof. Choose $z_n \in \Omega_n$. The condition $\operatorname{diam}(\Omega_n) \to 0$ says precisely that $\{z_n\}$ is Cauchy, so it converges to some w. Since each Ω_n is compact and contains z_m for $m \geq n$, $w \in \Omega_n$ for all n. If $w' \neq w$ had the same property, then $|w - w'| > 0$ would violate $\operatorname{diam}(\Omega_n) \to 0$. ■

An open set Ω is **connected** if it cannot be written as $\Omega = \Omega_1 \cup \Omega_2$ with Ω_1, Ω_2 disjoint non-empty open sets. A connected open set in $\mathbb{C}$ is a **region**. A closed set F is connected if it cannot be

written as $F = F_1 \cup F_2$ with F_1, F_2 disjoint non-empty closed sets. Equivalently, an open set Ω is connected iff any two points of Ω can be joined by a curve in Ω.

2 Functions on the complex plane

2.1 Continuous functions

f defined on Ω is **continuous** at $z_0 \in \Omega$ if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(z) - f(z_0)| < \epsilon$ whenever $z \in \Omega$, $|z - z_0| < \delta$; equivalently, $z_n \to z_0$ implies $f(z_n) \to f(z_0)$. f is continuous on Ω if continuous at every point. Sums and products of continuous functions are continuous. A function of $z = x + iy$ is continuous iff it is continuous as a function of the two real variables x, y. If f is continuous, so is $z \mapsto |f(z)|$. f **attains a maximum** at z_0 if $|f(z)| \leq |f(z_0)|$ for all $z \in \Omega$ (reversed for a minimum).

Theorem 2.1

A continuous function on a compact set Ω is bounded and attains a maximum and minimum on Ω.

(The proof is the same as for functions of a real variable.)

2.2 Holomorphic functions

Definition (holomorphic function)

Let Ω be open and f complex-valued on Ω. f is **holomorphic at** $z_0 \in \Omega$ if the quotient

$$\frac{f(z_0 + h) - f(z_0)}{h} \tag{1}$$

converges to a limit as $h \to 0$ ($h \in \mathbb{C}$, $h \neq 0$, $z_0 + h \in \Omega$). The limit is the **derivative** $f'(z_0)$. f is **holomorphic on** Ω if it is holomorphic at every point of Ω; if C is closed, f is holomorphic on C if it is holomorphic on some open set containing C; f holomorphic on all of $\mathbb{C}$ is **entire**.

In the limit h may approach 0 from any direction. The terms *regular* or *complex differentiable* are also used. Holomorphic functions turn out to be infinitely complex differentiable and *analytic* (they have power series expansions), in contrast with real-differentiable functions.

Example 1

$f(z) = z$ is holomorphic on any open set with $f'(z) = 1$; any polynomial $p(z) = a_0 + a_1 z + \cdots + a_n z^n$ is entire, with $p'(z) = a_1 + \cdots + n a_n z^{n-1}$ (by Proposition 2.2).

Example 2

$1/z$ is holomorphic on any open set not containing 0, with $f'(z) = -1/z^2$.

Example 3

$f(z) = \overline{z}$ is not holomorphic: $\dfrac{f(z_0 + h) - f(z_0)}{h} = \dfrac{\overline{h}}{h}$, which has no limit as $h \to 0$ (it is 1 for

real h and -1 for purely imaginary h).

Clearly f is holomorphic at z_0 iff there is $a \in \mathbb{C}$ with

$$f(z_0+h) - f(z_0) - ah = h\psi(h), \tag{2}$$

where ψ is defined for small h and $\lim_{h\to 0}\psi(h) = 0$; then $a = f'(z_0)$. Hence f is continuous wherever it is holomorphic. Arguing as for one real variable, using (2):

Proposition 2.2

If f and g are holomorphic in Ω, then:
(i) $f+g$ is holomorphic in Ω and $(f+g)' = f'+g'$.
(ii) fg is holomorphic in Ω and $(fg)' = f'g + fg'$.
(iii) If $g(z_0) \neq 0$, then f/g is holomorphic at z_0 and $(f/g)' = \dfrac{f'g - fg'}{g^2}$. Moreover, if $f : \Omega \to U$ and $g : U \to \mathbb{C}$ are holomorphic, the chain rule holds: $(g\circ f)'(z) = g'(f(z))f'(z)$ for all $z \in \Omega$.

Proof [Supplement](chain rule). Write $f(z+h)-f(z) = f'(z)h + h\psi_1(h)$ and $g(w+k)-g(w) = g'(w)k + k\psi_2(k)$ with $\psi_j \to 0$, $w = f(z)$. With $k = f(z+h)-f(z) \to 0$, $g(f(z+h)) - g(f(z)) = g'(w)k + k\psi_2(k) = g'(w)f'(z)h + h[g'(w)\psi_1(h) + (f'(z)+\psi_1(h))\psi_2(k)]$, and the bracket tends to 0. (i)–(iii) are similar. ■

Complex-valued functions as mappings. $\overline{z}$ corresponds to $F : (x,y) \mapsto (x,-y)$, which is linear, hence real-differentiable to all orders; so real differentiability does not imply holomorphy. To $f = u + iv$ associate $F(x,y) = (u(x,y), v(x,y))$. F is **differentiable** at $P_0 = (x_0,y_0)$ if there is a linear $J : \mathbb{R}^2 \to \mathbb{R}^2$ with

$$\frac{|F(P_0+H) - F(P_0) - J(H)|}{|H|} \to 0 \quad \text{as } |H| \to 0,\ H \in \mathbb{R}^2; \tag{3}$$

then J is the Jacobian matrix $J_F = \begin{pmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{pmatrix}$. If f is holomorphic at z_0, taking $h = h_1$ real in (1) gives $f'(z_0) = \frac{\partial f}{\partial x}(z_0)$, and taking $h = ih_2$ purely imaginary gives $f'(z_0) = \frac{1}{i}\frac{\partial f}{\partial y}(z_0)$. Hence $\frac{\partial f}{\partial x} = \frac{1}{i}\frac{\partial f}{\partial y}$, and separating real and imaginary parts (using $1/i = -i$) gives the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Define the operators

$$\frac{\partial}{\partial z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + \frac{1}{i}\frac{\partial}{\partial y}\Big), \qquad \frac{\partial}{\partial \overline{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} - \frac{1}{i}\frac{\partial}{\partial y}\Big).$$

Proposition 2.3

If f is holomorphic at z_0, then

$$\frac{\partial f}{\partial \overline{z}}(z_0) = 0 \qquad \text{and} \qquad f'(z_0) = \frac{\partial f}{\partial z}(z_0) = 2\frac{\partial u}{\partial z}(z_0).$$

Also, if we write $F(x,y) = f(z)$, then F is differentiable in the sense of real variables, and $\det J_F(x_0,y_0) = |f'(z_0)|^2$.

Proof. Taking real and imaginary parts, the Cauchy–Riemann equations are equivalent to $\partial f/\partial \overline{z} = 0$. By the observation above, $f'(z_0) = \frac{1}{2}(\frac{\partial f}{\partial x}(z_0) + \frac{1}{i}\frac{\partial f}{\partial y}(z_0)) = \frac{\partial f}{\partial z}(z_0)$, and the Cauchy–Riemann equations give $\partial f/\partial z = 2\partial u/\partial z$. For differentiability of F: if $H = (h_1, h_2)$ and $h = h_1 + ih_2$, the Cauchy–Riemann equations imply

$$J_F(x_0, y_0)(H) = \Big(\frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}\Big)(h_1 + ih_2) = f'(z_0)h,$$

identifying complex numbers with pairs of reals. Finally, by the Cauchy–Riemann equations,

$$\det J_F(x_0, y_0) = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial v}{\partial x}\frac{\partial u}{\partial y} = \Big(\frac{\partial u}{\partial x}\Big)^2 + \Big(\frac{\partial u}{\partial y}\Big)^2 = \Big|2\frac{\partial u}{\partial z}\Big|^2 = |f'(z_0)|^2. \tag{4}$$

■

The converse:

Theorem 2.4

Suppose $f = u + iv$ is a complex-valued function defined on an open set Ω. If u and v are continuously differentiable and satisfy the Cauchy–Riemann equations on Ω, then f is holomorphic on Ω and $f'(z) = \partial f/\partial z$.

Proof. Write

$$u(x+h_1, y+h_2) - u(x, y) = \frac{\partial u}{\partial x}h_1 + \frac{\partial u}{\partial y}h_2 + |h|\psi_1(h), \qquad v(x+h_1, y+h_2) - v(x, y) = \frac{\partial v}{\partial x}h_1 + \frac{\partial v}{\partial y}h_2 + |h|\psi_2(h),$$

with $\psi_j(h) \to 0$ as $|h| \to 0$, $h = h_1 + ih_2$ (differentiability follows from continuity of the partials). Using the Cauchy–Riemann equations,

$$f(z + h) - f(z) = \Big(\frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}\Big)(h_1 + ih_2) + |h|\psi(h),$$

with $\psi = \psi_1 + i\psi_2 \to 0$. So f is holomorphic and $f'(z) = 2\partial u/\partial z = \partial f/\partial z$. ■

2.3 Power series

The complex exponential is $e^z = \sum_{n=0}^{\infty} z^n/n!$. For real z this is the usual exponential, and the series converges absolutely for every $z \in \mathbb{C}$, since $|z^n/n!| = |z|^n/n!$ and $\sum |z|^n/n! = e^{|z|} < \infty$; the estimate shows uniform convergence on every disc. We will see that e^z is entire and its derivative is obtained termwise: $(e^z)' = \sum_{n=0}^{\infty} nz^{n-1}/n! = \sum_{m=0}^{\infty} z^m/m! = e^z$. The geometric series $\sum z^n$ converges absolutely only in $|z| < 1$, with sum $1/(1 - z)$ (holomorphic on $\mathbb{C} - \{1\}$), since $\sum_{n=0}^{N} z^n = \frac{1-z^{N+1}}{1-z}$ and $z^{N+1} \to 0$ for $|z| < 1$.

A **power series** is an expansion $\sum_{n=0}^{\infty} a_n z^n$ with $a_n \in \mathbb{C}$ (5). If it converges absolutely at z_0, it converges for all $|z| \le |z_0|$.

Theorem 2.5

Given a power series $\sum_{n=0}^{\infty} a_n z^n$, there exists $0 \le R \le \infty$ such that:
(i) If $|z| < R$ the series converges absolutely.
(ii) If $|z| > R$ the series diverges. Moreover, if we use the convention that $1/0 = \infty$ and $1/\infty = 0$, then R is given by Hadamard's formula

$$1/R = \limsup |a_n|^{1/n}.$$

R is the **radius of convergence** and $|z| < R$ the **disc of convergence**; $R = \infty$ for e^z and $R = 1$ for the geometric series.

Proof. Let $L = 1/R$ and suppose $L \neq 0, \infty$ (these cases are easy). If $|z| < R$, choose $\epsilon > 0$ with $(L+\epsilon)|z| = r < 1$. By definition of L, $|a_n|^{1/n} \leq L + \epsilon$ for large n, so $|a_n||z|^n \leq \{(L+\epsilon)|z|\}^n = r^n$, and comparison with $\sum r^n$ gives convergence. If $|z| > R$, a similar argument gives a sequence of terms whose absolute values tend to infinity, so the series diverges. ■

Remark

On the boundary $|z| = R$ either convergence or divergence can occur.

Further entire power series are

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \qquad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!},$$

agreeing with the usual functions for real z, and related to the exponential by the **Euler formulas**

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Theorem 2.6

The power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ defines a holomorphic function in its disc of convergence. The derivative of f is also a power series obtained by differentiating term by term the series for f, that is, $f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}$. Moreover, f' has the same radius of convergence as f.

Proof. Since $\lim n^{1/n} = 1$, $\limsup |a_n|^{1/n} = \limsup |n a_n|^{1/n}$, so $\sum a_n z^n$ and $\sum n a_n z^n$, hence $\sum n a_n z^{n-1}$, have the same radius of convergence (Hadamard's formula).

Let $g(z) = \sum n a_n z^{n-1}$, let R be the radius of convergence, and $|z_0| < r < R$. Write $f(z) = S_N(z) + E_N(z)$ with $S_N(z) = \sum_{n=0}^{N} a_n z^n$, $E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n$. If $|z_0 + h| < r$,

$$\frac{f(z_0+h) - f(z_0)}{h} - g(z_0) = \Big(\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\Big) + (S_N'(z_0) - g(z_0)) + \frac{E_N(z_0+h) - E_N(z_0)}{h}.$$

Since $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$,

$$\Big|\frac{E_N(z_0+h) - E_N(z_0)}{h}\Big| \leq \sum_{n=N+1}^{\infty} |a_n| \Big|\frac{(z_0+h)^n - z_0^n}{h}\Big| \leq \sum_{n=N+1}^{\infty} |a_n| n r^{n-1},$$

the tail of a convergent series (as g converges absolutely on $|z| < R$). Given $\epsilon > 0$ choose N_1 so that $N > N_1$ makes this $< \epsilon$, and N_2 so that $N > N_2$ gives $|S_N'(z_0) - g(z_0)| < \epsilon$. Fix $N > \max(N_1, N_2)$; since the derivative of a polynomial is obtained termwise, there is $\delta > 0$ with $|\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)| < \epsilon$ for $|h| < \delta$. Then $|\frac{f(z_0+h)-f(z_0)}{h} - g(z_0)| < 3\epsilon$ for $|h| < \delta$. ■

Corollary 2.7

A power series is infinitely complex differentiable in its disc of convergence, and the higher derivatives are also power series obtained by termwise differentiation.

A power series centered at z_0 is $f(z) = \sum a_n (z - z_0)^n$; with $g(w) = \sum a_n w^n$, $f(z) = g(z - z_0)$, so everything about g transfers, e.g. $f'(z) = \sum n a_n (z - z_0)^{n-1}$.

Definition (analytic function)

f defined on an open set Ω is **analytic** (has a power series expansion) at $z_0 \in \Omega$ if there is a power series $\sum a_n(z-z_0)^n$ with positive radius of convergence such that $f(z) = \sum a_n(z-z_0)^n$ for z near z_0. If this holds at every point of Ω, f is analytic on Ω.

By Theorem 2.6, analytic functions are holomorphic; the converse (Chapter 2) justifies using the two terms interchangeably.

3 Integration along curves

A **parametrized curve** is a function $z(t)$ from a closed interval $[a,b] \subset \mathbb{R}$ to $\mathbb{C}$. It is **smooth** if $z'(t)$ exists and is continuous on $[a,b]$ and $z'(t) \neq 0$; at $t = a, b$, z' means the one-sided derivatives

$$z'(a) = \lim_{h\to 0,\, h>0} \frac{z(a+h)-z(a)}{h}, \qquad z'(b) = \lim_{h\to 0,\, h<0} \frac{z(b+h)-z(b)}{h}.$$

It is **piecewise-smooth** if z is continuous on $[a,b]$ and smooth on intervals $[a_k, a_{k+1}]$ for some $a = a_0 < a_1 < \cdots < a_n = b$ (one-sided derivatives at the a_k may differ). Two parametrizations $z : [a,b] \to \mathbb{C}$, $\tilde{z} : [c,d] \to \mathbb{C}$ are **equivalent** if there is a continuously differentiable bijection $s \mapsto t(s)$ of $[c,d]$ onto $[a,b]$ with $t'(s) > 0$ and $\tilde{z}(s) = z(t(s))$; $t' > 0$ means orientation is preserved. The equivalence class determines a smooth curve $\gamma \subset \mathbb{C}$, the image of $[a,b]$ with the orientation given by z. γ^- is γ with orientation reversed, parametrized e.g. by $z^-(t) = z(b+a-t)$. The **end-points** $z(a)$, $z(b)$ do not depend on the parametrization; γ begins at $z(a)$ and ends at $z(b)$. γ is **closed** if $z(a) = z(b)$, and **simple** if $z(t) \neq z(s)$ unless $s = t$ (for closed curves: unless $s = t$, or $s = a$ and $t = b$). For brevity, a *curve* means a piecewise-smooth curve.

The circle $C_r(z_0)$ has the **positive orientation** (counterclockwise) given by $z(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$, and the negative orientation given by $z(t) = z_0 + re^{-it}$. C denotes a general positively oriented circle.

Definition (integral along a curve; length)

For a smooth curve γ parametrized by $z : [a,b] \to \mathbb{C}$ and f continuous on γ,

$$\int_\gamma f(z)\,dz = \int_a^b f(z(t))z'(t)\,dt.$$

For piecewise-smooth γ, $\int_\gamma f(z)\,dz = \sum_{k=0}^{n-1} \int_{a_k}^{a_{k+1}} f(z(t))z'(t)\,dt$. The **length** of a smooth curve is $\text{length}(\gamma) = \int_a^b |z'(t)|\,dt$, and that of a piecewise-smooth curve is the sum of the lengths of its smooth parts.

These are independent of the parametrization: if $\tilde{z}$ is equivalent to z, by change of variables and the chain rule

$$\int_a^b f(z(t))z'(t)\,dt = \int_c^d f(z(t(s)))z'(t(s))t'(s)\,ds = \int_c^d f(\tilde{z}(s))\tilde{z}'(s)\,ds.$$

Proposition 3.1

Integration of continuous functions over curves satisfies the following properties:
(i) It is linear: if $\alpha, \beta \in \mathbb{C}$, then $\int_\gamma (\alpha f(z) + \beta g(z))\,dz = \alpha \int_\gamma f(z)\,dz + \beta \int_\gamma g(z)\,dz$.
(ii) If γ^- is γ with the reverse orientation, then $\int_\gamma f(z)\,dz = -\int_{\gamma^-} f(z)\,dz$.
(iii) One has the inequality $\left|\int_\gamma f(z)\,dz\right| \leq \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$.

Proof. (i) follows from linearity of the Riemann integral.

(ii) [Supplement]: with $z^-(t) = z(a+b-t)$, $\int_a^b f(z^-(t))(z^-)'(t)\,dt = -\int_a^b f(z(a+b-t))z'(a+b-t)\,dt = -\int_a^b f(z(s))z'(s)\,ds$. (iii): $\left|\int_\gamma f(z)\,dz\right| \le \sup_{t\in[a,b]}|f(z(t))| \int_a^b |z'(t)|\,dt \le \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$. ■

A **primitive** for f on Ω is a function F holomorphic on Ω with $F'(z) = f(z)$ for all $z \in \Omega$.

Theorem 3.2

If a continuous function f has a primitive F in Ω, and γ is a curve in Ω that begins at w_1 and ends at w_2, then $\int_\gamma f(z)\,dz = F(w_2) - F(w_1)$.

Proof. If γ is smooth with parametrization $z : [a, b] \to \mathbb{C}$, $z(a) = w_1$, $z(b) = w_2$, then by the chain rule and the fundamental theorem of calculus

$$\int_\gamma f(z)\,dz = \int_a^b F'(z(t))z'(t)\,dt = \int_a^b \frac{d}{dt}F(z(t))\,dt = F(z(b)) - F(z(a)).$$

If γ is piecewise-smooth we get a telescoping sum: $\sum_{k=0}^{n-1}[F(z(a_{k+1})) - F(z(a_k))] = F(z(b)) - F(z(a))$. ■

Corollary 3.3

If γ is a closed curve in an open set Ω, and f is continuous and has a primitive in Ω, then $\oint_\gamma f(z)\,dz = 0$.

This is immediate since the end-points of a closed curve coincide. For example, $1/z$ has no primitive in $\mathbb{C} - \{0\}$: for the unit circle C, $z(t) = e^{it}$,

$$\oint_C f(z)\,dz = \int_0^{2\pi} \frac{ie^{it}}{e^{it}}\,dt = 2\pi i \neq 0.$$

This innocent calculation lies at the heart of the theory.

Corollary 3.4

If f is holomorphic in a region Ω and $f' = 0$, then f is constant.

Proof. Fix $w_0 \in \Omega$; it suffices to show $f(w) = f(w_0)$ for all $w \in \Omega$. Since Ω is connected, there is a curve γ joining w_0 to w. f is a primitive of f', so $\int_\gamma f'(z)\,dz = f(w) - f(w_0)$ by Theorem 3.2. But $f' = 0$, so the integral is 0 and $f(w) = f(w_0)$. ■

Chapter 2. Cauchy's Theorem and Its Applications

Overview

Loosely, Cauchy's theorem says that if f is holomorphic in an open set Ω and $\gamma \subset \Omega$ is a closed curve whose interior is also in Ω, then

$$\int_\gamma f(z)\,dz = 0. \tag{1}$$

A precise general formulation requires defining the "interior" of a curve unambiguously, which is not easy; at this stage we restrict to regions bounded by "toy contours", whose interiors are obvious. The starting point: it suffices that f have a primitive (Corollary 3.3), and primitives exist by *Goursat's theorem*: $\int_T f(z)\,dz = 0$ for a triangle T whose interior lies in Ω. This leads to the *Cauchy integral formula*

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad (z \text{ inside } C),$$

and hence to the regularity of holomorphic functions (existence of derivatives of all orders—decisively false for real variables), power series expansions, analytic continuation, Liouville's theorem and the fundamental theorem of algebra, and Morera's theorem.

1 Goursat's theorem

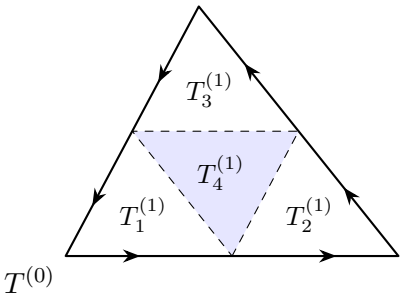


Figure. *Goursat's subdivision: joining the midpoints splits $T^{(0)}$ into four triangles, each oriented like $T^{(0)}$. The dashed interior edges are traversed twice in opposite directions and cancel, so $\int_{\partial T^{(0)}} f = \sum_{j=1}^{4} \int_{\partial T_j^{(1)}} f$, and one of the four integrals is at least a quarter of the total in absolute value.*

By Corollary 3.3, if f has a primitive in Ω then $\int_\gamma f = 0$ for closed $\gamma \subset \Omega$. Conversely, if the integral vanishes for certain curves, a primitive will exist. Goursat's theorem is the starting point from which most other results of this chapter are deduced. (Its interest is that its proof needs only the existence of f' at each point, not its continuity.)

Theorem 1.1

If Ω is an open set in $\mathbb{C}$, and $T \subset \Omega$ a triangle whose interior is also contained in Ω, then

$$\int_T f(z)\,dz = 0$$

whenever f is holomorphic in Ω.

Proof. Let $T^{(0)}$ be the original triangle (positively oriented), with diameter $d^{(0)}$ and perimeter $p^{(0)}$. Bisect each side and connect the midpoints, creating four similar triangles $T_1^{(1)}, \dots, T_4^{(1)}$, oriented

consistently with $T^{(0)}$. After cancellation of the integrals over interior sides traversed in opposite directions,

$$\int_{T^{(0)}} f(z)\,dz = \int_{T_1^{(1)}} f + \int_{T_2^{(1)}} f + \int_{T_3^{(1)}} f + \int_{T_4^{(1)}} f. \tag{2}$$

For some j, $\left|\int_{T^{(0)}} f\right| \le 4\left|\int_{T_j^{(1)}} f\right|$, for otherwise (2) would be contradicted. Call this triangle $T^{(1)}$; its diameter and perimeter are $d^{(1)} = \frac{1}{2}d^{(0)}$, $p^{(1)} = \frac{1}{2}p^{(0)}$. Repeating, we obtain triangles $T^{(0)}, T^{(1)}, \dots, T^{(n)}, \dots$ with

$$\Big|\int_{T^{(0)}} f(z)\,dz\Big| \le 4^n \Big|\int_{T^{(n)}} f(z)\,dz\Big|, \qquad d^{(n)} = 2^{-n}d^{(0)}, \quad p^{(n)} = 2^{-n}p^{(0)}.$$

Let $\mathcal{T}^{(n)}$ be the solid closed triangle with boundary $T^{(n)}$. These form a nested sequence of compact sets with diameters tending to 0, so by Proposition 1.4 there is a unique point z_0 in all of them. Since f is holomorphic at z_0,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0), \qquad \psi(z) \to 0 \text{ as } z \to z_0.$$

The constant $f(z_0)$ and the linear function $f'(z_0)(z - z_0)$ have primitives, so by Corollary 3.3

$$\int_{T^{(n)}} f(z)\,dz = \int_{T^{(n)}} \psi(z)(z - z_0)\,dz. \tag{3}$$

z_0 is in the solid triangle $\mathcal{T}^{(n)}$ and z on its boundary, so $|z - z_0| \le d^{(n)}$, and by Proposition 3.1(iii)

$$\Big|\int_{T^{(n)}} f(z)\,dz\Big| \le \epsilon_n d^{(n)} p^{(n)} = \epsilon_n 4^{-n} d^{(0)} p^{(0)}, \qquad \epsilon_n = \sup_{z \in T^{(n)}} |\psi(z)| \to 0.$$

Hence $\left|\int_{T^{(0)}} f\right| \le 4^n \left|\int_{T^{(n)}} f\right| \le \epsilon_n d^{(0)} p^{(0)}$, and letting $n \to \infty$ concludes the proof. ∎

Corollary 1.2

If f is holomorphic in an open set Ω that contains a rectangle R and its interior, then $\int_R f(z)\,dz = 0$.

Proof. Split R by a diagonal into two triangles T_1, T_2, oriented consistently; then $\int_R f = \int_{T_1} f + \int_{T_2} f = 0$. ∎

2 Local existence of primitives and Cauchy's theorem in a disc

Motivation. Goursat's theorem handles triangles. To pass to arbitrary closed curves in a disc we construct a primitive F with $F' = f$ by integrating along polygonal paths; then $\int_\gamma f = F(\text{end}) - F(\text{start}) = 0$ for every closed γ.

Theorem 2.1

A holomorphic function in an open disc has a primitive in that disc.

Proof. After a translation we may assume the disc D is centered at 0. For $z \in D$ let γ_z be the polygonal line (at most two segments) from 0 horizontally to $\tilde{z} = \mathrm{Re}(z)$ and then vertically to z, oriented from 0 to z, and define

$$F(z) = \int_{\gamma_z} f(w)\,dw.$$

Fix $z \in D$ and h so small that $z+h \in D$. In $F(z+h)-F(z) = \int_{\gamma_{z+h}} f - \int_{\gamma_z} f$, the integrals over the common segment from the origin cancel; completing a square and a triangle and applying Goursat's theorem for triangles and rectangles, we are left with

$$F(z+h)-F(z) = \int_{\eta} f(w)\,dw,$$

η the segment from z to $z+h$. Since f is continuous at z, $f(w) = f(z)+\psi(w)$ with $\psi(w) \to 0$ as $w \to z$. Hence

$$F(z+h)-F(z) = f(z)\int_{\eta} dw + \int_{\eta} \psi(w)\,dw. \tag{4}$$

The constant 1 has primitive w, so the first integral is h (Theorem 3.2); and $\left|\int_{\eta}\psi(w)\,dw\right| \le \sup_{w\in\eta}|\psi(w)|\,|h|$. The supremum tends to 0 as $h \to 0$, so (4) gives $\lim_{h\to 0}\frac{F(z+h)-F(z)}{h} = f(z)$. ■

So every holomorphic function has a primitive *locally*. The theorem holds not only for discs but for other sets as well ("toy contours" below).

Theorem 2.2 (Cauchy's theorem for a disc)

If f is holomorphic in a disc, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in that disc.

Proof. f has a primitive (Theorem 2.1); apply Corollary 3.3. ■

Corollary 2.3

Suppose f is holomorphic in an open set containing the circle C and its interior. Then $\int_C f(z)\,dz = 0$.

Proof. Let D be the disc bounded by C. There is a slightly larger disc $D' \supset \overline{D}$ on which f is holomorphic; apply Cauchy's theorem in D'. ■

Definition (toy contour)

A **toy contour** is any closed curve whose interior is obvious and for which a construction as in Theorem 2.1 is possible in a neighborhood of the curve and its interior. Its **positive orientation** is the one for which the interior is to the left as we travel along it.

Circles, triangles and rectangles are toy contours. Another important one is the **keyhole** Γ: two almost complete circles, one large and one small, connected by a narrow corridor. If f is holomorphic in a neighborhood of Γ and its interior, it is holomorphic inside a slightly larger keyhole Γ' whose interior contains $\overline{\operatorname{int}\Gamma}$. Fix $z_0 \in \operatorname{int}\Gamma'$; for $z \in \operatorname{int}\Gamma'$ let γ_z be a curve in $\operatorname{int}\Gamma'$ from z_0 to z made of finitely many horizontal or vertical segments. For any other such curve $\tilde{\gamma}_z$, Corollary 1.2 gives $\int_{\gamma_z} f = \int_{\tilde{\gamma}_z} f$, so $F(z) = \int_{\gamma_z} f$ is well defined, and as above F is a primitive of f; hence $\int_\Gamma f = 0$. In general, for a toy contour γ,

$$\int_\gamma f(z)\,dz = 0$$

whenever f is holomorphic in an open set containing γ and its interior. Further examples used later: the rectangular keyhole, the multiple keyhole, the semicircle, the indented semicircle, the sector and the parallelogram. For general curves, Appendix B proves the Jordan curve theorem for piecewise-smooth curves: a simple closed piecewise-smooth curve has a well-defined "simply connected" interior, and Cauchy's theorem holds in that setting.

3 Evaluation of some integrals

We show how some integrals may be evaluated using Cauchy's theorem; a systematic approach via residues is in Chapter 3.

Example 1

If $\xi \in \mathbb{R}$, then

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx, \tag{5}$$

i.e. $e^{-\pi x^2}$ is its own Fourier transform. For $\xi = 0$ this is the known integral $1 = \int_{-\infty}^{\infty} e^{-\pi x^2}\, dx$. Suppose $\xi > 0$ and let $f(z) = e^{-\pi z^2}$, entire. Let γ_R be the rectangle with vertices R, $R + i\xi$, $-R + i\xi$, $-R$, positively oriented. By Cauchy's theorem

$$\int_{\gamma_R} f(z)\, dz = 0. \tag{6}$$

The integral over the real segment is $\int_{-R}^{R} e^{-\pi x^2}\, dx \to 1$. The integral over the right vertical side is $I(R) = \int_0^{\xi} f(R+iy)\, i\, dy = \int_0^{\xi} e^{-\pi(R^2+2iRy-y^2)}\, i\, dy$, and $|I(R)| \le Ce^{-\pi R^2} \to 0$ (ξ fixed); similarly for the left side. The top side contributes

$$\int_R^{-R} e^{-\pi(x+i\xi)^2}\, dx = -e^{\pi\xi^2} \int_{-R}^{R} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx.$$

Letting $R \to \infty$, (6) gives $0 = 1 - e^{\pi\xi^2} \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx$. If $\xi < 0$ use the symmetric rectangle in the lower half-plane. (This *shifting of the contour* has many applications.)

Example 2

$$\int_0^{\infty} \frac{1 - \cos x}{x^2}\, dx = \frac{\pi}{2}.$$

Let $f(z) = (1 - e^{iz})/z^2$ and integrate over the indented semicircle in the upper half-plane: the segments $[-R, -\epsilon]$ and $[\epsilon, R]$, the small semicircle γ_ϵ^+ of radius ϵ (negatively oriented) and the large one γ_R^+ (positively oriented). By Cauchy's theorem

$$\int_{-R}^{-\epsilon} \frac{1 - e^{ix}}{x^2}\, dx + \int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2}\, dz + \int_{\epsilon}^{R} \frac{1 - e^{ix}}{x^2}\, dx + \int_{\gamma_R^+} \frac{1 - e^{iz}}{z^2}\, dz = 0.$$

Since $|(1-e^{iz})/z^2| \le 2/|z|^2$ in the upper half-plane, the integral over γ_R^+ tends to 0 as $R \to \infty$, so $\int_{|x|\ge\epsilon} \frac{1-e^{ix}}{x^2}\, dx = -\int_{\gamma_\epsilon^+} \frac{1-e^{iz}}{z^2}\, dz$. Now $f(z) = \frac{-iz}{z^2} + E(z)$ with E bounded near 0, and on γ_ϵ^+, $z = \epsilon e^{i\theta}$, $dz = i\epsilon e^{i\theta} d\theta$, so

$$\int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2}\, dz \to \int_{\pi}^{0} (-i)(i)\, d\theta = -\pi \quad \text{as } \epsilon \to 0.$$

Taking real parts, $\int_{-\infty}^{\infty} \frac{1-\cos x}{x^2}\, dx = \pi$; the integrand is even.

4 Cauchy's integral formulas

Motivation. Cauchy's theorem says integrals of holomorphic functions over closed curves vanish; applied to $f(\zeta)/(\zeta - z)$ it says much more, that the values of f inside a disc are determined by

its values on the boundary circle. Differentiating under the integral then shows that holomorphic functions are infinitely differentiable and equal to their Taylor series, which is the key difference from real analysis.

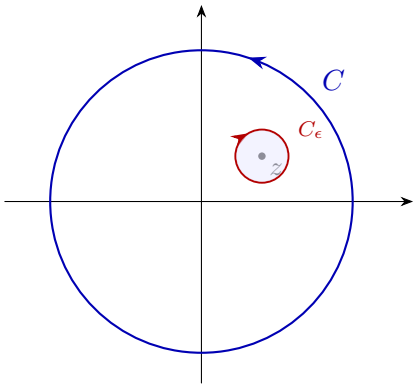


Figure. *For z inside C, $\zeta \mapsto f(\zeta)/(\zeta - z)$ is holomorphic between C and a small circle C_ϵ around z, so both circles give the same integral. As $\epsilon \to 0$ the small integral tends to $2\pi i f(z)$, giving $f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta$.*

Integral representations recover a function on a large set from its values on a smaller set (cf. the Poisson integral formula for the steady-state heat equation in the disc).

Theorem 4.1

Suppose f is holomorphic in an open set containing the closure of a disc D. If C denotes the boundary circle of this disc with the positive orientation, then

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad \text{for any point } z \in D.$$

Proof. Fix $z \in D$ and consider the keyhole $\Gamma_{\delta,\epsilon}$ omitting z: the circle C and a small circle of radius ϵ about z, joined by a corridor of width δ. $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic away from $\zeta = z$, so $\int_{\Gamma_{\delta,\epsilon}} F(\zeta)\,d\zeta = 0$ by Cauchy's theorem for this toy contour. Letting $\delta \to 0$, continuity of F makes the integrals over the two sides of the corridor cancel, leaving C (positively oriented) and the small circle C_ϵ about z (negatively oriented). Write

$$F(\zeta) = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}. \tag{8}$$

The first term is bounded (as f is holomorphic), so its integral over C_ϵ tends to 0 as $\epsilon \to 0$. And

$$\int_{C_\epsilon} \frac{f(z)}{\zeta - z}\,d\zeta = -f(z)\int_0^{2\pi} \frac{\epsilon i e^{-it}}{\epsilon e^{-it}}\,dt = -f(z)2\pi i.$$

So in the limit $0 = \int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta - 2\pi i f(z)$. ∎

Remark

The same holds for other toy contours: if f is holomorphic in an open set containing a positively oriented rectangle R and its interior, then $f(z) = \frac{1}{2\pi i}\int_R \frac{f(\zeta)}{\zeta - z}\,d\zeta$ for z in the interior of R (use a rectangular keyhole). The integral vanishes for z outside R, since then $f(\zeta)/(\zeta - z)$ is holomorphic inside R.

Corollary 4.2

If f is holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, if $C \subset \Omega$ is a circle whose interior is also contained in Ω, then

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta$$

for all z in the interior of C.

Proof. Induction on n; $n = 0$ is the Cauchy integral formula. Suppose f has up to $n-1$ complex derivatives with $f^{(n-1)}(z) = \frac{(n-1)!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^n}\, d\zeta$. For small h,

$$\frac{f^{(n-1)}(z+h) - f^{(n-1)}(z)}{h} = \frac{(n-1)!}{2\pi i} \int_C f(\zeta) \frac{1}{h} \Big[\frac{1}{(\zeta - z - h)^n} - \frac{1}{(\zeta - z)^n} \Big] d\zeta. \tag{9}$$

Since $A^n - B^n = (A-B)[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}]$, with $A = 1/(\zeta - z - h)$, $B = 1/(\zeta - z)$ the bracket in (9) is

$$\frac{h}{(\zeta - z - h)(\zeta - z)} [A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

For small h, $z + h$ and z stay at a finite distance from C, so as $h \to 0$ the quotient converges to

$$\frac{(n-1)!}{2\pi i} \int_C f(\zeta) \frac{1}{(\zeta - z)^2} \frac{n}{(\zeta - z)^{n-1}}\, d\zeta = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta.$$

■

The formulas of Theorem 4.1 and Corollary 4.2 are the **Cauchy integral formulas**.

Corollary 4.3 (Cauchy inequalities)

If f is holomorphic in an open set that contains the closure of a disc D centered at z_0 and of radius R, then

$$|f^{(n)}(z_0)| \leq \frac{n!\, \|f\|_C}{R^n},$$

where $\|f\|_C = \sup_{z \in C} |f(z)|$ denotes the supremum of $|f|$ on the boundary circle C.

Proof. By the Cauchy integral formula,

$$|f^{(n)}(z_0)| = \Big| \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta \Big| = \frac{n!}{2\pi} \Big| \int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Rie^{i\theta}\, d\theta \Big| \leq \frac{n!}{2\pi} \frac{\|f\|_C}{R^n} 2\pi.$$

■

Theorem 4.4

Suppose f is holomorphic in an open set Ω. If D is a disc centered at z_0 and whose closure is contained in Ω, then f has a power series expansion at z_0

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D$, and the coefficients are given by $a_n = \dfrac{f^{(n)}(z_0)}{n!}$ for all $n \geq 0$.

Proof. Fix $z \in D$. By the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta, \tag{10}$$

C the boundary of D. Write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)}. \tag{11}$$

Since $\zeta \in C$ and $z \in D$ is fixed, there is $0 < r < 1$ with $\left|\frac{z - z_0}{\zeta - z_0}\right| < r$, so

$$\frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)} = \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0}\right)^n, \tag{12}$$

uniformly for $\zeta \in C$. Interchanging sum and integral in (10)–(12),

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta\right)(z - z_0)^n.$$

The formula for a_n follows from the Cauchy integral formulas for derivatives (or by differentiating the series). ∎

This gives another proof that holomorphic functions are indefinitely differentiable. The expansion at z_0 converges in *any* disc centered at z_0 whose closure lies in Ω; in particular an entire function has a power series expansion about 0 converging on all of $\mathbb{C}$.

Corollary 4.5 (Liouville's theorem)

If f is entire and bounded, then f is constant.

Proof. It suffices to show $f' = 0$ ($\mathbb{C}$ is connected; Corollary 3.4). For each $z_0 \in \mathbb{C}$ and $R > 0$ the Cauchy inequalities give $|f'(z_0)| \le B/R$, B a bound for f. Let $R \to \infty$. ∎

Corollary 4.6

Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_0$ with complex coefficients has a root in $\mathbb{C}$.

Proof. If P has no roots, $1/P$ is a bounded entire function. Indeed, with $a_n \neq 0$, $\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n}\right)$ for $z \neq 0$; the parenthesis tends to 0 as $|z| \to \infty$, so with $c = |a_n|/2$ there is R with $|P(z)| \ge c|z|^n$ for $|z| > R$. P is continuous with no roots in $|z| \le R$, so it is bounded below there as well. By Liouville's theorem $1/P$ is constant, contradicting that P is non-constant. ∎

Corollary 4.7

Every polynomial $P(z) = a_n z^n + \cdots + a_0$ of degree $n \ge 1$ has precisely n roots in $\mathbb{C}$. If these roots are denoted by $w_1, \ldots, w_n$, then P can be factored as $P(z) = a_n(z - w_1)(z - w_2)\cdots(z - w_n)$.

Proof. P has a root w_1. Writing $z = (z - w_1) + w_1$ and expanding, $P(z) = b_n(z - w_1)^n + \cdots + b_1(z - w_1) + b_0$ with $b_n = a_n$; $P(w_1) = 0$ forces $b_0 = 0$, so $P(z) = (z - w_1)Q(z)$ with Q of degree $n - 1$. By induction $P(z) = c(z - w_1)\cdots(z - w_n)$, and comparing coefficients of z^n, $c = a_n$. ∎

Finally, *analytic continuation*: the "genetic code" of a holomorphic function is determined by its values on arbitrarily small sets. Here Ω must be connected.

Theorem 4.8

Suppose f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points with a limit point in Ω. Then f is identically 0.

Proof. Let $z_0 \in \Omega$ be a limit point of $\{w_k\}$ with $f(w_k) = 0$. First, $f \equiv 0$ near z_0: on a disc $D \subset \Omega$ centered at z_0, $f(z) = \sum a_n(z - z_0)^n$. If $f \not\equiv 0$ on D, let m be the smallest integer with $a_m \neq 0$; then $f(z) = a_m(z - z_0)^m(1 + g(z - z_0))$ with $g(z - z_0) \to 0$ as $z \to z_0$. Taking $z = w_k \neq z_0$ close to z_0, $a_m(w_k - z_0)^m \neq 0$ and $1 + g(w_k - z_0) \neq 0$, contradicting $f(w_k) = 0$.

Now let U be the interior of $\{z \in \Omega : f(z) = 0\}$; U is open and non-empty. U is closed in Ω: if $z_n \in U$ and $z_n \to z \in \Omega$, then $f(z) = 0$ by continuity, and f vanishes near z by the argument just given, so $z \in U$. With $V = \Omega - U$, U and V are disjoint open sets with $\Omega = U \cup V$; Ω is connected and $z_0 \in U$, so $U = \Omega$. ■

Corollary 4.9

Suppose f and g are holomorphic in a region Ω and $f(z) = g(z)$ for all z in some non-empty open subset of Ω (or more generally for z in some sequence of distinct points with limit point in Ω). Then $f(z) = g(z)$ throughout Ω.

If f, F are analytic in regions $\Omega \subset \Omega'$ and agree on Ω, F is an **analytic continuation** of f into Ω'; by the corollary it is unique.

5 Further applications

Motivation. The integral formula turns many analytic questions into estimates of integrals. We collect the consequences used repeatedly later: a converse of Cauchy's theorem (Morera), the stability of holomorphy under uniform limits and under integration in a parameter, the reflection principle, and approximation by polynomials and rational functions.

5.1 Morera's theorem

Theorem 5.1 (Morera)

Suppose f is a continuous function in the open disc D such that for any triangle T contained in D, $\int_T f(z)\,dz = 0$. Then f is holomorphic.

Proof. By the proof of Theorem 2.1, f has a primitive F in D with $F' = f$. By regularity (Corollary 4.2), F is indefinitely (hence twice) complex differentiable, so f is holomorphic. ■

5.2 Sequences of holomorphic functions

Theorem 5.2

If $\{f_n\}_{n=1}^{\infty}$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω, then f is holomorphic in Ω.

Proof. Let D be a disc with $\overline{D} \subset \Omega$ and T a triangle in D. By Goursat, $\int_T f_n = 0$ for all n. Since $f_n \to f$ uniformly on $\overline{D}$, f is continuous and $\int_T f_n \to \int_T f$, so $\int_T f = 0$, and by Morera's theorem f is holomorphic in D. This holds for every such D. ■

This fails for real variables: by Weierstrass's theorem every continuous function on $[0,1]$ is a uniform limit of polynomials, yet not every continuous function is differentiable.

Theorem 5.3

Under the hypotheses of the previous theorem, the sequence of derivatives $\{f_n'\}_{n=1}^{\infty}$ converges uniformly to f' on every compact subset of Ω.

Proof. We may assume the convergence is uniform on all of Ω. For $\delta > 0$ let $\Omega_\delta = \{z \in \Omega : \overline{D_\delta(z)} \subset \Omega\}$, the points at distance $> \delta$ from the boundary. It suffices to show $f_n' \to f'$ uniformly on each Ω_δ, which follows by applying to $F = f_n - f$ the inequality

$$\sup_{z\in\Omega_\delta} |F'(z)| \le \frac{1}{\delta} \sup_{\zeta\in\Omega} |F(\zeta)|, \tag{13}$$

valid for F holomorphic in Ω. For $z \in \Omega_\delta$, $\overline{D_\delta(z)} \subset \Omega$ and by the Cauchy integral formula

$$|F'(z)| = \Big|\frac{1}{2\pi i}\int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta - z)^2}\, d\zeta\Big| \le \frac{1}{2\pi} \sup_{\zeta\in\Omega}|F(\zeta)| \frac{1}{\delta^2} 2\pi\delta = \frac{1}{\delta}\sup_{\zeta\in\Omega}|F(\zeta)|.$$

■

Likewise $f_n^{(k)} \to f^{(k)}$ uniformly on compact subsets for every $k \ge 0$. In practice Theorem 5.2 constructs holomorphic functions as series $F(z) = \sum_{n=1}^{\infty} f_n(z)$ (14) converging uniformly on compact subsets of Ω, e.g. the zeta function (Chapter 6).

5.3 Holomorphic functions defined in terms of integrals

After a linear change of variables we take the interval $[0,1]$.

Theorem 5.4

Let $F(z,s)$ be defined for $(z,s) \in \Omega \times [0,1]$ where Ω is an open set in $\mathbb{C}$. Suppose F satisfies the following properties:
(i) $F(z,s)$ is holomorphic in z for each s.
(ii) F is continuous on $\Omega \times [0,1]$. Then the function f defined on Ω by $f(z) = \int_0^1 F(z,s)\, ds$ is holomorphic.

Proof. Instead of justifying an interchange of integrals in Morera's theorem, view the integral as a uniform limit of Riemann sums. Let $f_n(z) = \frac{1}{n}\sum_{k=1}^{n} F(z, k/n)$, holomorphic in Ω by (i). On a disc D with $\overline{D} \subset \Omega$, F is uniformly continuous on $\overline{D} \times [0,1]$, so given $\epsilon > 0$ there is $\delta > 0$ with $\sup_{z\in\overline{D}} |F(z,s_1) - F(z,s_2)| < \epsilon$ when $|s_1 - s_2| < \delta$. For $n > 1/\delta$ and $z \in D$,

$$|f_n(z) - f(z)| = \Big|\sum_{k=1}^{n} \int_{(k-1)/n}^{k/n} [F(z,k/n) - F(z,s)]\, ds\Big| \le \sum_{k=1}^{n} \frac{\epsilon}{n} = \epsilon.$$

So $f_n \to f$ uniformly on D, and f is holomorphic in D by Theorem 5.2; hence in Ω. ■

5.4 Schwarz reflection principle

Holomorphic functions are rigid: there are holomorphic functions in a disc, continuous on its closure, that cannot be continued beyond it; and they vanish identically if they vanish on a line segment. Yet there is a simple and useful extension phenomenon. Let Ω be open and symmetric with respect to the real line ($z \in \Omega \iff \overline{z} \in \Omega$), Ω^+ its part in the upper half-plane, Ω^- the part in the lower half-plane, and $I = \Omega \cap \mathbb{R}$; then $\Omega = \Omega^+ \cup I \cup \Omega^-$.

Theorem 5.5 (Symmetry principle)

If f^+ and f^- are holomorphic functions in Ω^+ and Ω^- respectively, that extend continuously to I and $f^+(x) = f^-(x)$ for all $x \in I$, then the function f defined on Ω by

$$f(z) = \begin{cases} f^+(z) & z \in \Omega^+, \\ f^+(z) = f^-(z) & z \in I, \\ f^-(z) & z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω.

Proof. f is continuous on Ω; the issue is holomorphy at points of I. Let $D \subset \Omega$ be a disc centered at a point of I; we apply Morera's theorem. If a triangle $T \subset D$ does not meet I, $\int_T f = 0$ since f is holomorphic in the half-discs. If one side or vertex of T lies on I and the rest is, say, in the upper half-disc, let T_ϵ be obtained by raising that edge or vertex by ϵ; $\int_{T_\epsilon} f = 0$ as T_ϵ lies in the upper half-disc, and letting $\epsilon \to 0$, by continuity $\int_T f = 0$. If the interior of T meets I, write T as a union of triangles each with an edge or vertex on I. By Morera's theorem f is holomorphic in D. ■

Theorem 5.6 (Schwarz reflection principle)

Suppose that f is a holomorphic function in Ω^+ that extends continuously to I and such that f is real-valued on I. Then there exists a function F holomorphic in all of Ω such that $F = f$ on Ω^+.

Proof. Define $F(z) = \overline{f(\overline{z})}$ for $z \in \Omega^-$. If $z, z_0 \in \Omega^-$ then $\overline{z}, \overline{z_0} \in \Omega^+$ and $f(\overline{z}) = \sum a_n(\overline{z} - \overline{z_0})^n$ near $\overline{z_0}$, so $F(z) = \sum \overline{a_n}(z - z_0)^n$ and F is holomorphic in Ω^-. Since f is real on I, $\overline{f(x)} = f(x)$ for $x \in I$, so F extends continuously up to I and agrees with f there. Apply the symmetry principle. ■

5.5 Runge's approximation theorem

By Weierstrass's theorem, continuous functions on a compact interval are uniform limits of polynomials. What conditions on a compact $K \subset \mathbb{C}$ ensure that every function holomorphic in a neighborhood of K is a uniform limit of polynomials on K? For a disc this follows from power series. Some condition is necessary: for $f(z) = 1/z$ on the unit circle $K = C$, $\int_C f = 2\pi i$ while $\int_C p = 0$ for every polynomial p, so uniform approximation is impossible. The right condition is that K^c be connected (a variant of this example shows it is also necessary).

Theorem 5.7 (Runge)

Any function holomorphic in a neighborhood of a compact set K can be approximated uniformly on K by rational functions whose singularities are in K^c. If K^c is connected, any function holomorphic in a neighborhood of K can be approximated uniformly on K by polynomials.

The second part follows from the first by "pushing the singularities to infinity". The key is an integral representation from the Cauchy formula for squares:

Lemma 5.8

Suppose f is holomorphic in an open set Ω, and $K \subset \Omega$ is compact. Then there exist finitely

many segments $\gamma_1, \dots, \gamma_N$ in $\Omega - K$ such that

$$f(z) = \sum_{n=1}^{N} \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z}\, d\zeta \qquad \text{for all } z \in K. \tag{15}$$

Proof. Let $d = c \cdot d(K, \Omega^c)$ with $c < 1/\sqrt{2}$, and consider a grid of closed squares with sides of length d parallel to the axes. Let $\mathcal{Q} = \{Q_1, \dots, Q_M\}$ be the squares of the grid meeting K, with positively oriented boundaries ∂Q_m, and let $\gamma_1, \dots, \gamma_N$ be the sides of squares in $\mathcal{Q}$ that do not belong to two adjacent squares of $\mathcal{Q}$. By the choice of d, each γ_n lies in Ω and does not meet K (if it did, it would belong to two adjacent squares of $\mathcal{Q}$). For $z \in K$ not on the boundary of a square of $\mathcal{Q}$, $z \in Q_j$ for some j, and by Cauchy's theorem and formula

$$\frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta = \begin{cases} f(z) & m = j, \\ 0 & m \neq j, \end{cases} \qquad \text{so} \qquad f(z) = \sum_{m=1}^{M} \frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

The integrals over a common side of two adjacent squares are taken once in each direction and cancel; this gives (15) for such z. Since the γ_n lie in K^c, continuity extends (15) to all $z \in K$. ∎

Lemma 5.9

For any line segment γ entirely contained in $\Omega - K$, there exists a sequence of rational functions with singularities on γ that approximate the integral $\int_\gamma f(\zeta)/(\zeta - z)\, d\zeta$ uniformly on K.

Proof. With a parametrization $\gamma(t)$, $t \in [0,1]$, $\int_\gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta = \int_0^1 \frac{f(\gamma(t))\gamma'(t)}{\gamma(t) - z}\, dt$. Since γ does not meet K, the integrand $F(z,t)$ is jointly continuous on the compact set $K \times [0,1]$, hence uniformly continuous, and as in the proof of Theorem 5.4 the Riemann sums of $\int_0^1 F(z,t)\, dt$ approximate it uniformly on K. Each Riemann sum is a rational function with singularities on γ. ∎

Lemmas 5.8 and 5.9 prove the first part of Theorem 5.7. A rational function whose only singularity is z_0 is a polynomial in $1/(z - z_0)$, so the second part follows from:

Lemma 5.10

If K^c is connected and $z_0 \notin K$, then the function $1/(z - z_0)$ can be approximated uniformly on K by polynomials.

Proof. Choose z_1 outside a large open disc D centered at 0 containing K. Then

$$\frac{1}{z - z_1} = -\frac{1}{z_1} \frac{1}{1 - z/z_1} = -\sum_{n=0}^{\infty} \frac{z^n}{z_1^{n+1}},$$

uniformly for $z \in K$; the partial sums are polynomials approximating $1/(z - z_1)$ uniformly on K, and hence any power $1/(z - z_1)^k$ is a uniform limit of polynomials on K. It now suffices to approximate $1/(z - z_0)$ uniformly on K by polynomials in $1/(z - z_1)$. Since K^c is connected, there is a curve $\gamma \subset K^c$, $\gamma(0) = z_0$, $\gamma(1) = z_1$. Let $\rho = \frac{1}{2} d(K, \gamma) > 0$ (both are compact), and choose points $w_0 = z_0, w_1, \dots, w_\ell = z_1$ on γ with $|w_j - w_{j+1}| < \rho$.

Claim: if $w, w' \in \gamma$ with $|w - w'| < \rho$, then $1/(z - w)$ is a uniform limit on K of polynomials in $1/(z - w')$. Indeed

$$\frac{1}{z - w} = \frac{1}{z - w'} \cdot \frac{1}{1 - \frac{w - w'}{z - w'}} = \sum_{n=0}^{\infty} \frac{(w - w')^n}{(z - w')^{n+1}},$$

and the series converges uniformly for $z \in K$ since $|z - w'| \geq d(K, \gamma) = 2\rho$ gives $|w - w'|/|z - w'| < \rho/(2\rho) = 1/2$. Applying the claim successively, $1/(z - w_0)$ is a uniform limit of polynomials in $1/(z - w_1)$, hence (as powers of $1/(z - w_1)$ are limits of polynomials in $1/(z - w_2)$) of polynomials in $1/(z - w_2)$, and so on, up to polynomials in $1/(z - w_\ell) = 1/(z - z_1)$. ∎

Chapter 3. Meromorphic Functions and the Logarithm

Overview

A general principle, implicit in Riemann's work: analytic functions are essentially characterized by their singularities—globally, holomorphic functions are "effectively" determined by their zeros, and meromorphic functions by their zeros and poles. We study isolated singularities, in order of increasing severity: *removable singularities*, *poles*, and *essential singularities*. Poles lead to the *calculus of residues*, whose key ingredient is $\int_C dz/z = 2\pi i$. Indefinite integrals of functions with singularities, like $\int dz/z = \log z$, may fail to be single-valued; exploiting this gives the *argument principle*, which counts zeros inside a curve and yields the open mapping theorem and the maximum principle. To define the logarithm properly we introduce homotopies and simply connected domains.

1 Zeros and poles

A **point singularity** (or **isolated singularity**) of f is a point z_0 such that f is defined in a neighborhood of z_0 but not at z_0. For $f(z) = z$ on the punctured plane, 0 is removable (set $f(0) = 0$). $g(z) = 1/z$ cannot be extended continuously to 0; it grows to infinity, and 0 is a pole. $h(z) = e^{1/z}$ grows indefinitely as $z \to 0$ along the positive reals, tends to 0 along the negative reals, and oscillates rapidly but stays bounded along the imaginary axis: removable singularities and poles do not tell the whole story.

z_0 is a **zero** of f if $f(z_0) = 0$. By analytic continuation, the zeros of a non-trivial holomorphic function are isolated.

Theorem 1.1

Suppose that f is holomorphic in a connected open set Ω, has a zero at a point $z_0 \in \Omega$, and does not vanish identically in Ω. Then there exists a neighborhood $U \subset \Omega$ of z_0, a non-vanishing holomorphic function g on U, and a unique positive integer n such that $f(z) = (z - z_0)^n g(z)$ for all $z \in U$.

Proof. Since Ω is connected and $f \not\equiv 0$, f is not identically zero near z_0 (Theorem 4.8). Near z_0, $f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k$; let n be the smallest integer with $a_n \neq 0$. Then $f(z) = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z)$, with g holomorphic and non-vanishing near z_0 (as $a_n \neq 0$).

Uniqueness: if also $f(z) = (z - z_0)^m h(z)$ with $h(z_0) \neq 0$ and $m > n$, then $g(z) = (z - z_0)^{m-n} h(z)$ and letting $z \to z_0$ gives $g(z_0) = 0$, a contradiction; $m < n$ similarly forces $h(z_0) = 0$. So $m = n$ and $h = g$. ∎

f has a **zero of order** n (multiplicity n) at z_0; a zero of order 1 is **simple**. A **deleted neighborhood** of z_0 is $\{z : 0 < |z - z_0| < r\}$. A function f defined in a deleted neighborhood of z_0 has a **pole** at z_0 if $1/f$, defined to be zero at z_0, is holomorphic in a full neighborhood of z_0.

Theorem 1.2

If f has a pole at $z_0 \in \Omega$, then in a neighborhood of that point there exist a non-vanishing holomorphic function h and a unique positive integer n such that $f(z) = (z - z_0)^{-n} h(z)$.

Proof. By Theorem 1.1, $1/f(z) = (z - z_0)^n g(z)$ with g holomorphic and non-vanishing near z_0; take $h = 1/g$. ∎

n is the **order** of the pole (it describes the rate of growth); a pole of order 1 is **simple**.

Theorem 1.3

If f has a pole of order n at z_0, then

$$f(z) = \frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-n+1}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z-z_0} + G(z), \tag{1}$$

where G is a holomorphic function in a neighborhood of z_0.

Proof. Expand $h(z) = A_0 + A_1(z-z_0) + \cdots$ in Theorem 1.2; then $f(z) = (z-z_0)^{-n}(A_0 + A_1(z-z_0) + \cdots)$ has the stated form. ∎

The sum $\frac{a_{-n}}{(z-z_0)^n} + \cdots + \frac{a_{-1}}{z-z_0}$ is the **principal part** of f at z_0, and a_{-1} the **residue**, $\operatorname{res}_{z_0} f = a_{-1}$. All other terms of the principal part have primitives in a deleted neighborhood of z_0, so for the principal part P and any circle C centered at z_0, $\frac{1}{2\pi i}\int_C P(z)\,dz = a_{-1}$. For a simple pole, $\operatorname{res}_{z_0} f = \lim_{z\to z_0}(z-z_0)f(z)$.

Theorem 1.4

If f has a pole of order n at z_0, then

$$\operatorname{res}_{z_0} f = \lim_{z\to z_0} \frac{1}{(n-1)!}\Big(\frac{d}{dz}\Big)^{n-1}(z-z_0)^n f(z).$$

Proof. By (1), $(z-z_0)^n f(z) = a_{-n} + a_{-n+1}(z-z_0) + \cdots + a_{-1}(z-z_0)^{n-1} + G(z)(z-z_0)^n$; differentiate $n-1$ times and let $z \to z_0$. ∎

2 The residue formula

Motivation. If f has isolated singularities inside a contour, Cauchy's theorem no longer gives 0; instead each singularity contributes a residue, the coefficient of $(z-z_0)^{-1}$ in the Laurent expansion. This turns the computation of difficult real integrals into finding a few coefficients.

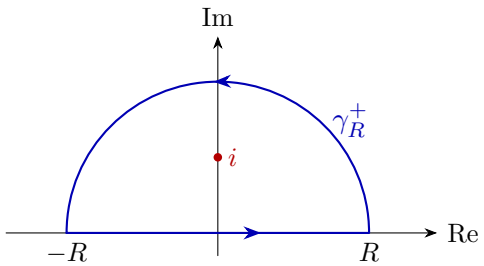


Figure. *The standard semicircular contour for $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$: only the pole at i lies inside, the arc contributes $O(1/R) \to 0$, and the residue formula gives $2\pi i \cdot \frac{1}{2i} = \pi$.*

Theorem 2.1

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C. Then

$$\int_C f(z)\,dz = 2\pi i \operatorname{res}_{z_0} f.$$

Proof. Using a keyhole contour avoiding the pole and letting the corridor width tend to 0, $\int_C f = \int_{C_\epsilon} f$, C_ϵ the small circle of radius ϵ about z_0. By the Cauchy integral formula applied to the constant

a_{-1}, $\frac{1}{2\pi i}\int_{C_\epsilon}\frac{a_{-1}}{z-z_0}\,dz = a_{-1}$; by the formulas for derivatives (Corollary 4.2), $\frac{1}{2\pi i}\int_{C_\epsilon}\frac{a_{-k}}{(z-z_0)^k}\,dz = 0$ for $k>1$; and $\int_{C_\epsilon} G = 0$ by Cauchy's theorem. By (1), $\int_{C_\epsilon} f = 2\pi i a_{-1}$. ■

Corollary 2.2

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for poles at the points $z_1,\dots,z_N$ inside C. Then

$$\int_C f(z)\,dz = 2\pi i\sum_{k=1}^{N}\operatorname{res}_{z_k} f.$$

Proof. Use a multiple keyhole with a loop around each pole; letting the corridor widths go to zero, the integral over C becomes a sum of integrals over small circles, to which Theorem 2.1 applies. ■

Corollary 2.3

Suppose that f is holomorphic in an open set containing a toy contour γ and its interior, except for poles at the points $z_1,\dots,z_N$ inside γ. Then

$$\int_\gamma f(z)\,dz = 2\pi i\sum_{k=1}^{N}\operatorname{res}_{z_k} f.$$

Here γ has positive orientation.

(Choose a keyhole appropriate to the toy contour and reduce to small circles around the poles.) This identity is the **residue formula**.

2.1 Examples

To compute $\int_{-\infty}^{\infty} f(x)\,dx$, extend f to $\mathbb{C}$ and choose toy contours γ_R with $\lim_{R\to\infty}\int_{\gamma_R} f(z)\,dz = \int_{-\infty}^{\infty} f(x)\,dx$; the residues give $\int_{\gamma_R} f$. Choosing γ_R is the challenge, often guided by the decay of f.

Example 1

$$\int_{-\infty}^{\infty}\frac{dx}{1+x^2} = \pi. \tag{2}$$

(Changing variables $x\mapsto x/y$, this says $\int P_y(x)\,dx = 1$ for the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$.) $f(z) = 1/(1+z^2) = 1/((z-i)(z+i))$ has simple poles at $\pm i$, and $\operatorname{res}_i f = 1/2i$. Let γ_R be the segment $[-R,R]$ together with the upper half-circle C_R^+ of radius R. For large R, $\int_{\gamma_R} f = 2\pi i\cdot\frac{1}{2i} = \pi$. Since $|f(z)|\le B/|z|^2$ on C_R^+, $\left|\int_{C_R^+} f\right| \le \frac{B}{R^2}\pi R \to 0$. Hence $\int_{-\infty}^{\infty} f(x)\,dx = \pi$. (The lower semicircle, with the other pole, works equally well.)

Example 2

An integral that plays an important role in Chapter 6:

$$\int_{-\infty}^{\infty}\frac{e^{ax}}{1+e^x}\,dx = \frac{\pi}{\sin\pi a}, \qquad 0<a<1.$$

Let $f(z) = e^{az}/(1+e^z)$ and γ_R the rectangle with sides on $\mathbb{R}$ and on $\operatorname{Im}(z) = 2\pi$, vertices $\pm R$, $\pm R + 2\pi i$. The only zero of the denominator inside is $z = \pi i$. There

$$(z-\pi i)f(z) = e^{az}\frac{z-\pi i}{e^z - e^{\pi i}}, \qquad \lim_{z\to\pi i}\frac{e^z - e^{\pi i}}{z-\pi i} = e^{\pi i} = -1,$$

so f has a simple pole at πi with residue $-e^{a\pi i}$, and

$$\int_{\gamma_R} f = -2\pi i e^{a\pi i}. \tag{3}$$

Let $I_R = \int_{-R}^{R} f(x)\,dx \to I$. The top side (right to left) contributes $-e^{2\pi i a} I_R$. On the right vertical side $A_R = \{R + it : 0 \le t \le 2\pi\}$, $\left|\int_{A_R} f\right| \le \int_0^{2\pi} \left|\frac{e^{a(R+it)}}{1+e^{R+it}}\right| dt \le Ce^{(a-1)R} \to 0$ since $a < 1$; the left side is bounded by $Ce^{-aR} \to 0$ since $a > 0$. So (3) gives $I - e^{2\pi i a} I = -2\pi i e^{a\pi i}$, whence

$$I = -2\pi i \frac{e^{a\pi i}}{1 - e^{2\pi i a}} = \frac{2\pi i}{e^{\pi i a} - e^{-\pi i a}} = \frac{\pi}{\sin \pi a}.$$

Example 3

$1/\cosh \pi x$ is its own Fourier transform: with $\cosh z = \frac{e^z + e^{-z}}{2}$,

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{\cosh \pi x}\,dx = \frac{1}{\cosh \pi \xi}.$$

Let $f(z) = e^{-2\pi i z\xi}/\cosh \pi z$ and γ_R the rectangle with vertices $\pm R$, $\pm R + 2i$. The denominator vanishes when $e^{2\pi z} = -1$; inside the rectangle the poles are $\alpha = i/2$ and $\beta = 3i/2$. Since

$$(z-\alpha)f(z) = 2e^{-2\pi i z\xi} e^{\pi z}\frac{z-\alpha}{e^{2\pi z} - e^{2\pi\alpha}}, \qquad \lim_{z\to\alpha}(z-\alpha)f(z) = 2e^{-2\pi i\alpha\xi}e^{\pi\alpha}\frac{1}{2\pi e^{2\pi\alpha}} = \frac{e^{\pi\xi}}{\pi i},$$

f has a simple pole at α with residue $e^{\pi\xi}/(\pi i)$; similarly at β with residue $-e^{3\pi\xi}/(\pi i)$. On $z = R + iy$, $0 \le y \le 2$: $|e^{-2\pi i z\xi}| \le e^{4\pi|\xi|}$ and $|\cosh \pi z| \ge \frac{1}{2}(e^{\pi R} - e^{-\pi R}) \to \infty$, so the vertical sides contribute 0 in the limit. Since $\cosh \pi z$ is periodic with period $2i$, the top side contributes $-e^{4\pi\xi} I$. The residue formula gives

$$I - e^{4\pi\xi} I = 2\pi i\Big(\frac{e^{\pi\xi}}{\pi i} - \frac{e^{3\pi\xi}}{\pi i}\Big) = -2e^{2\pi\xi}(e^{\pi\xi} - e^{-\pi\xi}),$$

and since $1 - e^{4\pi\xi} = -e^{2\pi\xi}(e^{2\pi\xi} - e^{-2\pi\xi})$,

$$I = 2\frac{e^{\pi\xi} - e^{-\pi\xi}}{e^{2\pi\xi} - e^{-2\pi\xi}} = \frac{2}{e^{\pi\xi} + e^{-\pi\xi}} = \frac{1}{\cosh \pi\xi}.$$

A similar argument gives, for $0 < a < 1$, with $\sinh z = (e^z - e^{-z})/2$,

$$\int_{-\infty}^{\infty} e^{-2\pi i x\xi}\frac{\sin \pi a}{\cosh \pi x + \cos \pi a}\,dx = \frac{2\sinh 2\pi a\xi}{\sinh 2\pi\xi};$$

$a = 1/2$ is the case above. This yields the Poisson kernel for the strip, and is used in the two-squares theorem (Chapter 10).

3 Singularities and meromorphic functions

Motivation. Isolated singularities come in three kinds, removable, poles and essential, and they behave very differently: near an essential singularity f comes arbitrarily close to every value (Casorati–Weierstrass). Classifying them tells us which functions can be treated as holomorphic functions with a few manageable poles.

Let f be holomorphic in Ω except possibly at $z_0 \in \Omega$. If f can be defined at z_0 so that it becomes holomorphic in all of Ω, z_0 is a **removable singularity**.

Theorem 3.1 (Riemann's theorem on removable singularities)

Suppose that f is holomorphic in an open set Ω except possibly at a point z_0 in Ω. If f is bounded on $\Omega - \{z_0\}$, then z_0 is a removable singularity.

Proof. Take a small disc D centered at z_0 with $\overline{D} \subset \Omega$, boundary circle C positively oriented. We show that for $z \in D$, $z \neq z_0$,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta. \tag{4}$$

By Theorem 5.4 the right side is holomorphic in all of D and agrees with f for $z \neq z_0$, giving the extension. To prove (4), fix z and use a multiple keyhole avoiding z and z_0; letting the corridors shrink,

$$\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = 0,$$

where γ_ϵ, γ'_ϵ are small negatively oriented circles of radius ϵ about z and z_0. As in the proof of the Cauchy integral formula, $\int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = -2\pi i f(z)$. Since f is bounded and $\zeta \in \gamma'_\epsilon$ stays away from z, $\left| \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta \right| \leq C\epsilon$. Let $\epsilon \to 0$. ∎

Corollary 3.2

Suppose that f has an isolated singularity at the point z_0. Then z_0 is a pole of f if and only if $|f(z)| \to \infty$ as $z \to z_0$.

Proof. If z_0 is a pole, $1/f$ has a zero at z_0, so $|f(z)| \to \infty$. Conversely, if $|f(z)| \to \infty$, then $1/f$ is bounded near z_0 and $1/|f(z)| \to 0$; so $1/f$ has a removable singularity at z_0 and vanishes there, i.e. z_0 is a pole. ∎

Isolated singularities fall into three categories: **removable** (f bounded near z_0), **poles** ($|f(z)| \to \infty$ as $z \to z_0$), and **essential singularities** (all others), e.g. $e^{1/z}$ at 0. Near an essential singularity a function typically behaves erratically:

Theorem 3.3 (Casorati–Weierstrass)

Suppose f is holomorphic in the punctured disc $D_r(z_0) - \{z_0\}$ and has an essential singularity at z_0. Then, the image of $D_r(z_0) - \{z_0\}$ under f is dense in the complex plane.

Proof. If not, there are $w \in \mathbb{C}$ and $\delta > 0$ with $|f(z) - w| > \delta$ on the punctured disc. Then $g(z) = 1/(f(z) - w)$ is holomorphic on the punctured disc and bounded by $1/\delta$, so it has a removable singularity at z_0 (Theorem 3.1). If $g(z_0) \neq 0$, $f(z) - w = 1/g(z)$ is holomorphic at z_0; if $g(z_0) = 0$, $f(z) - w$ has a pole at z_0. Both contradict that z_0 is essential. ∎

Picard proved much more: under these hypotheses f takes every complex value infinitely many times with at most one exception.

Definition (meromorphic function)

A function f on an open set Ω is **meromorphic** if there is a sequence $\{z_0, z_1, z_2, \dots\}$ with no limit points in Ω such that (i) f is holomorphic in $\Omega - \{z_0, z_1, \dots\}$, and (ii) f has poles at the points $z_0, z_1, \dots$.

If f is holomorphic for all large z, let $F(z) = f(1/z)$, holomorphic in a deleted neighborhood of 0. f has a **pole at infinity** if F has a pole at 0; similarly an **essential singularity at infinity**, or a **removable singularity** (holomorphic) at infinity. A meromorphic function in $\mathbb{C}$ that is holomorphic or has a pole at infinity is **meromorphic in the extended complex plane**. The principle stated at the outset appears here in its simplest form:

Theorem 3.4

The meromorphic functions in the extended complex plane are the rational functions.

Proof. $f(1/z)$ has a pole or removable singularity at 0, hence is holomorphic in a deleted neighborhood of 0; so f has only finitely many poles in the plane, say $z_1, \dots, z_n$. Near each z_k, $f(z) = f_k(z) + g_k(z)$ with f_k the principal part (a polynomial in $1/(z - z_k)$) and g_k holomorphic near z_k. Similarly $f(1/z) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z)$ with $\tilde{g}_\infty$ holomorphic near 0 and $\tilde{f}_\infty$ a polynomial in $1/z$; let $f_\infty(z) = \tilde{f}_\infty(1/z)$, a polynomial. Then $H = f - f_\infty - \sum_{k=1}^n f_k$ is entire (removable singularities at the z_k) and bounded ($H(1/z)$ is bounded near 0, as the principal part at ∞ was subtracted). By Liouville, H is constant, so f is rational. ∎

Consequently, a rational function is determined up to a multiplicative constant by the locations and multiplicities of its zeros and poles.

The Riemann sphere. In $\mathbb{R}^3$ with coordinates (X, Y, Z), identify the XY-plane with $\mathbb{C}$, and let $\mathbb{S}$ be the sphere of radius $1/2$ centered at $(0, 0, 1/2)$, with north pole $N = (0, 0, 1)$. For $W = (X, Y, Z) \in \mathbb{S} - \{N\}$, the line through N and W meets the XY-plane in a single point $w = x + iy$, the **stereographic projection** of W; this is a bijection between $\mathbb{S} - \{N\}$ and $\mathbb{C}$, given by

$$x = \frac{X}{1-Z}, \quad y = \frac{Y}{1-Z}; \qquad X = \frac{x}{x^2+y^2+1}, \quad Y = \frac{y}{x^2+y^2+1}, \quad Z = \frac{x^2+y^2}{x^2+y^2+1}.$$

As $|w| \to \infty$, $W \to N$, so N is the natural "point at infinity". The extended complex plane is thus the sphere $\mathbb{S}$, the **Riemann sphere** (the *one-point compactification* of $\mathbb{C}$). The point at infinity is on equal footing with the others, and a meromorphic function on the extended plane is a map of $\mathbb{S}$ to itself, sending poles to N.

4 The argument principle and applications

$\log f(z)$ is in general multiple-valued, but it must equal $\log|f(z)| + i \arg f(z)$, and its derivative $f'(z)/f(z)$ is single-valued; $\int_\gamma \frac{f'(z)}{f(z)}\,dz$ is the change in the argument of f along γ, determined (for closed γ) by the zeros and poles of f inside γ. While $\log(f_1 f_2) = \log f_1 + \log f_2$ fails in general, additivity holds for logarithmic derivatives:

$$\frac{(f_1 f_2)'}{f_1 f_2} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2}, \qquad \frac{\left(\prod_{k=1}^N f_k\right)'}{\prod_{k=1}^N f_k} = \sum_{k=1}^N \frac{f_k'}{f_k}.$$

If f has a zero of order n at z_0, $f(z) = (z - z_0)^n g(z)$ with g non-vanishing near z_0, so $\frac{f'(z)}{f(z)} = \frac{n}{z-z_0} + G(z)$, $G = g'/g$ holomorphic: f'/f has a simple pole with residue n. If f has a pole of order n, $f(z) = (z - z_0)^{-n} h(z)$ and $\frac{f'(z)}{f(z)} = \frac{-n}{z-z_0} + H(z)$. The residue formula gives:

Theorem 4.1 (Argument principle)

Suppose f is meromorphic in an open set containing a circle C and its interior. If f has no poles and never vanishes on C, then

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}\,dz = (\text{number of zeros of } f \text{ inside } C) - (\text{number of poles of } f \text{ inside } C),$$

where the zeros and poles are counted with their multiplicities.

Corollary 4.2

The above theorem holds for toy contours.

Three applications: Rouché's theorem (a continuity statement), the open mapping theorem, and the maximum modulus principle.

Theorem 4.3 (Rouché's theorem)

Suppose that f and g are holomorphic in an open set containing a circle C and its interior. If $|f(z)| > |g(z)|$ for all $z \in C$, then f and $f+g$ have the same number of zeros inside the circle C.

Proof. For $t \in [0,1]$ let $f_t(z) = f(z) + tg(z)$, so $f_0 = f$, $f_1 = f+g$, and let n_t be the number of zeros of f_t inside C (with multiplicities). $|f| > |g|$ on C implies f_t has no zeros on C, and by the argument principle $n_t = \frac{1}{2\pi i}\int_C \frac{f_t'(z)}{f_t(z)}\,dz$. $f_t'(z)/f_t(z)$ is jointly continuous in $t \in [0,1]$, $z \in C$ (numerator and denominator are, and the denominator does not vanish on C), so n_t is continuous in t. An integer-valued continuous function on $[0,1]$ is constant (intermediate value theorem), so $n_0 = n_1$. ■

A mapping is **open** if it maps open sets to open sets.

Theorem 4.4 (Open mapping theorem)

If f is holomorphic and non-constant in a region Ω, then f is open.

Proof. Let $w_0 = f(z_0)$; we show all w near w_0 are in the image. Write $g(z) = f(z) - w = (f(z) - w_0) + (w_0 - w) = F(z) + G(z)$. Choose $\delta > 0$ with $\{|z - z_0| \le \delta\} \subset \Omega$ and $f(z) \neq w_0$ on $|z - z_0| = \delta$ (zeros of $f - w_0$ are isolated), then $\epsilon > 0$ with $|f(z) - w_0| \ge \epsilon$ on $|z - z_0| = \delta$. If $|w - w_0| < \epsilon$, then $|F| > |G|$ on that circle, and by Rouché's theorem $g = F + G$ has a zero inside, since F has one (z_0). ■

The maximum of f in Ω means the maximum of $|f|$ in Ω.

Theorem 4.5 (Maximum modulus principle)

If f is a non-constant holomorphic function in a region Ω, then f cannot attain a maximum in Ω.

Proof. If f attained a maximum at z_0, then since f is open, the image $f(D)$ of a small disc D centered at z_0 is open and contains $f(z_0)$; so there are $z \in D$ with $|f(z)| > |f(z_0)|$. ■

Corollary 4.6

Suppose that Ω is a region with compact closure $\overline{\Omega}$. If f is holomorphic on Ω and continuous on $\overline{\Omega}$ then

$$\sup_{z\in\Omega}|f(z)| \le \sup_{z\in\overline{\Omega}-\Omega}|f(z)|.$$

Proof. $|f|$ attains its maximum on the compact set $\overline{\Omega}$, and not in Ω if f is non-constant; if f is constant the conclusion is trivial. ■

Remark

Compactness (boundedness) of $\overline{\Omega}$ is essential. Let Ω be the open first quadrant and $F(z) = e^{-iz^2}$. F is entire, $|F| = 1$ on both boundary half-lines $z = x$, $z = iy$, yet F is unbounded in Ω: $F(z) = e^{r^2}$ for $z = r\sqrt{i} = re^{i\pi/4}$.

5 Homotopies and simply connected domains

The key to the general Cauchy theorem and to multiple-valued functions is knowing in what regions a holomorphic function has a primitive—a global question (relevant to the logarithm, the primitive of $1/z$).

Definition (homotopic curves)

Let $\gamma_0, \gamma_1 \subset \Omega$ be curves with parametrizations on $[a, b]$ and common end-points $\gamma_0(a) = \gamma_1(a) = \alpha$, $\gamma_0(b) = \gamma_1(b) = \beta$. They are **homotopic in** Ω if for each $0 \le s \le 1$ there is a curve $\gamma_s \subset \Omega$, parametrized by $\gamma_s(t)$ on $[a, b]$, with $\gamma_s(a) = \alpha$, $\gamma_s(b) = \beta$ for every s, $\gamma_s(t)|_{s=0} = \gamma_0(t)$, $\gamma_s(t)|_{s=1} = \gamma_1(t)$, and $\gamma_s(t)$ jointly continuous in $s \in [0, 1]$, $t \in [a, b]$.

Loosely, one curve can be deformed into the other continuously without leaving Ω.

Theorem 5.1

If f is holomorphic in Ω, then $\int_{\gamma_0} f(z)\,dz = \int_{\gamma_1} f(z)\,dz$ whenever the two curves γ_0 and γ_1 are homotopic in Ω.

Proof. The key is: if two curves are close and have the same end-points, the integrals agree. $F(s,t) = \gamma_s(t)$ is continuous on $[0,1] \times [a,b]$; its image K is compact. There is $\epsilon > 0$ such that every disc of radius 3ϵ centered at a point of K lies in Ω: otherwise for every $\ell \ge 1$ there are $z_\ell \in K$, $w_\ell \in \Omega^c$ with $|z_\ell - w_\ell| < 1/\ell$; a subsequence $z_{\ell_k} \to z \in K$, then $w_{\ell_k} \to z$ and $z \in \Omega^c$ (Ω^c is closed), a contradiction. By uniform continuity of F, choose δ with $\sup_t|\gamma_{s_1}(t) - \gamma_{s_2}(t)| < \epsilon$ whenever $|s_1 - s_2| < \delta$.

Fix such s_1, s_2. Choose discs $D_0, \dots, D_n$ of radius 2ϵ and consecutive points $z_0, \dots, z_{n+1}$ on γ_{s_1}, $w_0, \dots, w_{n+1}$ on γ_{s_2}, such that the discs cover both curves and $z_i, z_{i+1}, w_i, w_{i+1} \in D_i$, with $z_0 = w_0$ the common beginning point and $z_{n+1} = w_{n+1}$ the common end-point. On each D_i let F_i be a primitive of f (Theorem 2.1). On $D_i \cap D_{i+1}$, F_i and F_{i+1} differ by a constant, so $F_{i+1}(z_{i+1}) - F_i(z_{i+1}) = F_{i+1}(w_{i+1}) - F_i(w_{i+1})$, i.e.

$$F_{i+1}(z_{i+1}) - F_{i+1}(w_{i+1}) = F_i(z_{i+1}) - F_i(w_{i+1}). \tag{5}$$

Hence

$$\int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f = \sum_{i=0}^{n}[F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^{n}[F_i(w_{i+1}) - F_i(w_i)] = \sum_{i=0}^{n}\left[F_i(z_{i+1}) - F_i(w_{i+1}) - (F_i(z_i) - F_i(w_i))\right],$$

which telescopes by (5) to $F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)) = 0$. Subdividing $[0,1]$ into intervals of length $< \delta$, finitely many applications take us from γ_0 to γ_1. ■

Definition (simply connected)

A region Ω is **simply connected** if any two curves in Ω with the same end-points are homotopic.

Example 1

A disc D is simply connected: $\gamma_s(t) = (1-s)\gamma_0(t) + s\gamma_1(t)$ lies on the segment joining $\gamma_0(t)$ and $\gamma_1(t)$, hence in D. The same works for rectangles and any open convex set.

Example 2

The slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ is simply connected: write $\gamma_j(t) = r_j(t)e^{i\theta_j(t)}$ with $r_j > 0$ and θ_j continuous, $|\theta_j(t)| < \pi$, and let $\gamma_s(t) = r_s(t)e^{i\theta_s(t)}$ with $r_s = (1-s)r_0 + sr_1$, $\theta_s = (1-s)\theta_0 + s\theta_1$; then $\gamma_s(t) \in \Omega$.

Example 3

With some effort one shows that the interior of a toy contour is simply connected, by dividing it into several subregions.

Example 4

The punctured plane $\mathbb{C} - \{0\}$ is not simply connected (proved after Corollary 5.3).

Theorem 5.2

Any holomorphic function in a simply connected domain has a primitive.

Proof. Fix $z_0 \in \Omega$ and define $F(z) = \int_\gamma f(w)\,dw$, the integral over any curve $\gamma \subset \Omega$ joining z_0 to z; this is independent of γ by Theorem 5.1, since Ω is simply connected. Then $F(z+h) - F(z) = \int_\eta f(w)\,dw$ with η the segment from z to $z+h$, and as in the proof of Theorem 2.1, $\lim_{h\to 0} \frac{F(z+h)-F(z)}{h} = f(z)$. ■

Corollary 5.3

If f is holomorphic in the simply connected region Ω, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in Ω.

That $\mathbb{C} - \{0\}$ is not simply connected now follows rigorously since $\int_C dz/z = 2\pi i \neq 0$ over the unit circle.

6 The complex logarithm

For $z = re^{i\theta}$ it is natural to set $\log z = \log r + i\theta$ ($\log r$ the standard logarithm of $r > 0$), but θ is unique only up to a multiple of 2π. Locally a continuous choice of θ is determined, but not globally:

winding once around the origin changes the logarithm by $2\pi i$. To obtain a single-valued function we must restrict the domain—the choice of a **branch** or **sheet** of the logarithm.

Theorem 6.1

Suppose that Ω is simply connected with $1 \in \Omega$, and $0 \notin \Omega$. Then in Ω there is a branch of the logarithm $F(z) = \log_\Omega(z)$ so that (i) F is holomorphic in Ω, (ii) $e^{F(z)} = z$ for all $z \in \Omega$, (iii) $F(r) = \log r$ whenever r is a real number and near 1.

So each branch $\log_\Omega(z)$ extends the standard logarithm.

Proof. $1/z$ is holomorphic in Ω; define $\log_\Omega(z) = F(z) = \int_\gamma \frac{dw}{w}$, γ any curve in Ω from 1 to z (path-independent as Ω is simply connected). As in Theorem 5.2, F is holomorphic with $F'(z) = 1/z$: (i). For (ii), $\frac{d}{dz}(ze^{-F(z)}) = e^{-F(z)} - zF'(z)e^{-F(z)} = (1 - zF'(z))e^{-F(z)} = 0$, so $ze^{-F(z)}$ is constant on the connected Ω (Corollary 3.4); at $z = 1$, $F(1) = 0$, so the constant is 1. For (iii), for real r near 1 integrate along the real segment: $F(r) = \int_1^r \frac{dx}{x} = \log r$. ∎

In the slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ we have the **principal branch** $\log z = \log r + i\theta$ for $z = re^{i\theta}$, $|\theta| < \pi$: integrate along the segment from 1 to r and the arc from r to z,

$$\log z = \int_1^r \frac{dx}{x} + \int_{\text{arc}} \frac{dw}{w} = \log r + \int_0^\theta \frac{ire^{it}}{re^{it}}\, dt = \log r + i\theta.$$

In general $\log(z_1 z_2) \neq \log z_1 + \log z_2$: for $z_1 = z_2 = e^{2\pi i/3}$, $\log z_1 = \log z_2 = 2\pi i/3$, while $z_1 z_2 = e^{-2\pi i/3}$ and $\log(z_1 z_2) = -2\pi i/3$. For the principal branch,

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = -\sum_{n=1}^{\infty} (-1)^n \frac{z^n}{n} \qquad \text{for } |z| < 1, \tag{6}$$

since both sides have derivative $1/(1+z)$ and vanish at $z = 0$.

For Ω simply connected with $1 \in \Omega$, $0 \notin \Omega$, and $\alpha \in \mathbb{C}$, define $z^\alpha = e^{\alpha \log z}$ (branch with $\log 1 = 0$). Then $1^\alpha = 1$, and for $\alpha = 1/n$, $(z^{1/n})^n = \prod_{k=1}^n e^{\frac{1}{n}\log z} = e^{\log z} = z$.

Theorem 6.2

If f is a nowhere vanishing holomorphic function in a simply connected region Ω, then there exists a holomorphic function g on Ω such that $f(z) = e^{g(z)}$.

g may be denoted $\log f(z)$; it determines a "branch" of that logarithm.

Proof. Fix $z_0 \in \Omega$ and let $g(z) = \int_\gamma \frac{f'(w)}{f(w)}\, dw + c_0$, γ any path in Ω from z_0 to z, with $e^{c_0} = f(z_0)$ (path-independent as Ω is simply connected). As in Theorem 2.1, g is holomorphic with $g' = f'/f$, and $\frac{d}{dz}(f(z)e^{-g(z)}) = 0$, so $f(z)e^{-g(z)}$ is constant; at z_0 it is $f(z_0)e^{-c_0} = 1$. ∎

7 Fourier series and harmonic functions

Motivation. Real and imaginary parts of holomorphic functions are harmonic, and on the unit circle a holomorphic power series becomes a Fourier series. This section uses these links to derive the mean value property for harmonic functions.

There is a simple and direct relation between Fourier series on the circle and power series of holomorphic functions in the disc. Suppose f is holomorphic in a disc $D_R(z_0)$, so $f(z) = \sum_{n=0}^\infty a_n (z - z_0)^n$ there.

Theorem 7.1

The coefficients of the power series expansion of f are given by

$$a_n = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta$$

for all $n \geq 0$ and $0 < r < R$. Moreover,

$$0 = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta$$

whenever $n < 0$.

Proof. Since $f^{(n)}(z_0) = a_n n!$, the Cauchy integral formula over the circle γ of radius r about z_0 gives, with $\zeta = z_0 + re^{i\theta}$,

$$a_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{(re^{i\theta})^{n+1}} rie^{i\theta}\, d\theta = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta.$$

For $n < 0$ the same computation gives $\frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta$, and since $-n > 0$, $f(\zeta)(\zeta - z_0)^{-n-1}$ is holomorphic in the disc; by Cauchy's theorem the integral vanishes. ■

So the Fourier coefficients of the restriction of f to a circle vanish for $n < 0$, and for $n \geq 0$ equal (up to r^n) the power series coefficients—another special characteristic of holomorphic functions. Since $a_0 = f(z_0)$:

Corollary 7.2 (Mean-value property)

If f is holomorphic in a disc $D_R(z_0)$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta \qquad \text{for any } 0 < r < R.$$

Corollary 7.3

If f is holomorphic in a disc $D_R(z_0)$, and $u = \mathrm{Re}(f)$, then

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta \qquad \text{for any } 0 < r < R.$$

This follows by taking real parts. Every harmonic function u in a disc is the real part of a holomorphic function there, so the mean-value property holds for all harmonic functions in a disc.

Chapter 4. The Fourier Transform

Overview

For f on $\mathbb{R}$ with suitable regularity and decay, the Fourier transform and its inversion are

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ix\xi}\,dx, \qquad f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi.$$

The theme of this chapter: the possibility of extending f to a holomorphic function is closely related to very rapid (e.g. exponential) decay of $\hat{f}$ at infinity. First, if f continues analytically to a horizontal strip and has moderate decrease, then $\hat{f}$ decays exponentially, and the inversion and Poisson summation formulas follow by contour integration. Second, assuming only Fourier inversion, the *Paley–Wiener theorem* says $\hat{f}$ is supported in $[-M, M]$ iff f extends to an entire function with $|f(z)| \le Ae^{2\pi M|z|}$ (*exponential type*). The decisive technique is shifting the contour (the real line) within a horizontal strip, exploiting that $e^{-2\pi iz\xi}$ decays or grows exponentially when $\operatorname{Im}(z) \neq 0$, according to the sign of $\xi \operatorname{Im}(z)$.

A function f is of **moderate decrease** if it is continuous and $|f(x)| \le A/(1+x^2)$ for some $A > 0$.

1 The class $\mathfrak{F}$

Book I proved Fourier inversion and Poisson summation when $|f(x)| \le \frac{A}{1+x^2}$ and $|\hat{f}(\xi)| \le \frac{A'}{1+\xi^2}$, a class containing e.g. the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$, $\hat{P}_y(\xi) = e^{-2\pi y|\xi|}$. Here we introduce a class suited to complex analysis.

Definition (the classes $\mathfrak{F}_a$ and $\mathfrak{F}$)

For $a > 0$, $\mathfrak{F}_a$ is the class of functions f such that:
(i) f is holomorphic in the horizontal strip $S_a = \{z \in \mathbb{C} : |\operatorname{Im}(z)| < a\}$;
(ii) there is $A > 0$ with $|f(x+iy)| \le \dfrac{A}{1+x^2}$ for all $x \in \mathbb{R}$ and $|y| < a$. $\mathfrak{F}$ is the class of functions belonging to $\mathfrak{F}_a$ for some a.

So $\mathfrak{F}_a$ consists of holomorphic functions on S_a of moderate decay on each horizontal line, uniformly in $-a < y < a$. Examples: $e^{-\pi z^2} \in \mathfrak{F}_a$ for all a; $\frac{1}{\pi}\frac{c}{c^2+z^2}$ (simple poles at $\pm ci$) belongs to $\mathfrak{F}_a$ for $0 < a < c$; $1/\cosh \pi z \in \mathfrak{F}_a$ for $|a| < 1/2$. By the Cauchy integral formulas, if $f \in \mathfrak{F}_a$ then $f^{(n)} \in \mathfrak{F}_b$ for $0 < b < a$. (The decrease $A/(1+x^2)$ may be weakened to $A/(1+|x|^{1+\epsilon})$ in most results below.)

2 Action of the Fourier transform on $\mathfrak{F}$

Motivation. For functions that extend holomorphically to a horizontal strip, contour shifting gives clean proofs of the Fourier inversion and Poisson summation formulas, without the convergence difficulties of the real-variable theory.

All three theorems are proved by contour integration.

Theorem 2.1

If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \le Be^{-2\pi b|\xi|}$ for any $0 \le b < a$.

Proof. $b = 0$ says $\hat{f}$ is bounded, clear from moderate decrease. Let $0 < b < a$ and $\xi > 0$. Shift the real line down by b: let $g(z) = f(z)e^{-2\pi iz\xi}$ and consider the rectangle with vertices $-R, R, R - ib, -R - ib$. The integral over the left vertical side is bounded by

$$\Big| \int_{-R-ib}^{-R} g(z)\,dz \Big| \le \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}|\,dt \le \int_0^b \frac{A}{R^2} e^{-2\pi t\xi}\,dt = O(1/R^2),$$

and similarly for the right side. By Cauchy's theorem, letting $R \to \infty$,

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - ib)e^{-2\pi i(x-ib)\xi}\,dx, \tag{1}$$

so $|\hat{f}(\xi)| \le \int_{-\infty}^{\infty} \frac{A}{1+x^2} e^{-2\pi b\xi}\,dx \le Be^{-2\pi b\xi}$. For $\xi < 0$ shift the real line up by b. ■

The further f extends (larger a), the better the decay of $\hat{f}$.

Theorem 2.2

If $f \in \mathfrak{F}$, then the Fourier inversion holds, namely

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi \qquad \text{for all } x \in \mathbb{R}.$$

Lemma 2.3

If A is positive and B is real, then $\int_0^\infty e^{-(A+iB)\xi}\,d\xi = \dfrac{1}{A+iB}$.

Proof. $|e^{-(A+iB)\xi}| = e^{-A\xi}$, so the integral converges, and $\int_0^R e^{-(A+iB)\xi}\,d\xi = \Big[-\frac{e^{-(A+iB)\xi}}{A+iB}\Big]_0^R \to \frac{1}{A+iB}$. ■

Proof of Theorem 2.2. Split $\int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi ix\xi}d\xi = \int_{-\infty}^{0} + \int_0^{\infty}$. Say $f \in \mathfrak{F}_a$, $0 < b < a$. By (1) and the lemma (the integrals converge absolutely),

$$\begin{aligned}\int_0^\infty \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi &= \int_0^\infty \int_{-\infty}^\infty f(u-ib)e^{-2\pi i(u-ib)\xi}e^{2\pi ix\xi}\,du\,d\xi = \int_{-\infty}^\infty f(u-ib)\int_0^\infty e^{-2\pi i(u-ib-x)\xi}\,d\xi\,du \\ &= \int_{-\infty}^\infty f(u-ib)\frac{1}{2\pi b + 2\pi i(u-x)}\,du = \frac{1}{2\pi i}\int_{-\infty}^\infty \frac{f(u-ib)}{u-ib-x}\,du = \frac{1}{2\pi i}\int_{L_1}\frac{f(\zeta)}{\zeta - x}\,d\zeta,\end{aligned}$$

$L_1 = \{u - ib : u \in \mathbb{R}\}$ traversed left to right. Similarly $\int_{-\infty}^0 \hat{f}(\xi)e^{2\pi ix\xi}d\xi = -\frac{1}{2\pi i}\int_{L_2}\frac{f(\zeta)}{\zeta - x}\,d\zeta$, L_2 the real line shifted up by b, left to right. Now for $x \in \mathbb{R}$ let γ_R be the rectangle with vertices $-R \pm ib$, $R \pm ib$. $f(\zeta)/(\zeta - x)$ has a simple pole at x with residue $f(x)$, so $f(x) = \frac{1}{2\pi i}\int_{\gamma_R}\frac{f(\zeta)}{\zeta - x}\,d\zeta$. As $R \to \infty$ the vertical sides contribute 0, and

$$f(x) = \frac{1}{2\pi i}\int_{L_1}\frac{f(\zeta)}{\zeta - x}\,d\zeta - \frac{1}{2\pi i}\int_{L_2}\frac{f(\zeta)}{\zeta - x}\,d\zeta = \int_0^\infty \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi + \int_{-\infty}^0 \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi.$$

■

Theorem 2.4 (Poisson summation formula)

If $f \in \mathfrak{F}$, then $\displaystyle\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat{f}(n)$.

Proof. Say $f \in \mathfrak{F}_a$, $0 < b < a$. $1/(e^{2\pi iz} - 1)$ has simple poles with residue $1/(2\pi i)$ at the integers, so $f(z)/(e^{2\pi iz} - 1)$ has simple poles at the integers n with residues $f(n)/2\pi i$. Apply the residue formula to the rectangle γ_N with vertices $\pm(N + \frac{1}{2}) \pm ib$:

$$\sum_{|n|\le N} f(n) = \int_{\gamma_N} \frac{f(z)}{e^{2\pi iz} - 1}\, dz.$$

Let $N \to \infty$; by moderate decrease the vertical sides contribute 0, and

$$\sum_{n\in\mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi iz} - 1}\, dz - \int_{L_2} \frac{f(z)}{e^{2\pi iz} - 1}\, dz, \tag{2}$$

L_1, L_2 the real line shifted down and up by b. If $|w| > 1$, $\frac{1}{w-1} = w^{-1}\sum_{n=0}^{\infty} w^{-n}$, so on L_1 (where $|e^{2\pi iz}| > 1$) $\frac{1}{e^{2\pi iz}-1} = e^{-2\pi iz}\sum_{n=0}^{\infty} e^{-2\pi inz}$; if $|w| < 1$, $\frac{1}{w-1} = -\sum_{n=0}^{\infty} w^n$, so on L_2 $\frac{1}{e^{2\pi iz}-1} = -\sum_{n=0}^{\infty} e^{2\pi inz}$. Substituting in (2) and shifting L_1, L_2 back to the real line (by (1) and its analogue),

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n=0}^{\infty}\int_{L_1} f(z)e^{-2\pi i(n+1)z}\, dz + \sum_{n=0}^{\infty}\int_{L_2} f(z)e^{2\pi inz}\, dz = \sum_{n=0}^{\infty}\hat{f}(n+1) + \sum_{n=0}^{\infty}\hat{f}(-n) = \sum_{n\in\mathbb{Z}}\hat{f}(n).$$

■

Consequences. Since $e^{-\pi x^2}$ is its own Fourier transform (Example 1), for $t > 0$ and $a \in \mathbb{R}$ the change of variables $x \mapsto t^{1/2}(x + a)$ shows that $f(x) = e^{-\pi t(x+a)^2}$ has $\hat{f}(\xi) = t^{-1/2}e^{-\pi\xi^2/t}e^{2\pi ia\xi}$. Poisson summation gives

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2}e^{-\pi n^2/t}e^{2\pi ina}. \tag{3}$$

For $a = 0$ this is the transformation law of the theta function $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$ $(t > 0)$:

$$\vartheta(t) = t^{-1/2}\vartheta(1/t) \qquad \text{for } t > 0, \tag{4}$$

used in Chapter 6 for the functional equation of ζ; the general case is used in Chapter 10 for the Jacobi theta function. Likewise, since $1/\cosh \pi x$ is its own Fourier transform (Example 3), $f(x) = e^{-2\pi iax}/\cosh(\pi x/t)$ has $\hat{f}(\xi) = t/\cosh(\pi(\xi + a)t)$, and

$$\sum_{n=-\infty}^{\infty} \frac{e^{-2\pi ian}}{\cosh(\pi n/t)} = \sum_{n=-\infty}^{\infty} \frac{t}{\cosh(\pi(n + a)t)}, \tag{5}$$

used in Chapter 10 for the two-squares theorem.

3 Paley–Wiener theorem

Motivation. Which functions have Fourier transforms supported in a bounded interval? Exactly those that extend to entire functions of exponential type. The theorem shows how decay on one side of the Fourier transform corresponds to analyticity on the other.

Now we assume no analyticity, but the validity of Fourier inversion when $|f(x)| \le A/(1 + x^2)$ and $|\hat{f}(\xi)| \le A'/(1 + \xi^2)$ (proved in Book I). A partial converse to Theorem 2.1:

Theorem 3.1

Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \le Ae^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : |\mathrm{Im}(z)| < b\}$, for any $0 < b < a$.

Proof. Let $f_n(z) = \int_{-n}^{n} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$, entire by Theorem 5.4. For $z \in S_b$ define $f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$; the integral converges absolutely, being majorized by $A\int e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi < \infty$ as $b < a$. Moreover for $z \in S_b$, $|f(z) - f_n(z)| \le A\int_{|\xi|\ge n} e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi \to 0$, so $f_n \to f$ uniformly in S_b, and f is holomorphic there by Theorem 5.2. ∎

Corollary 3.2

If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$, and f vanishes in a non-empty open interval, then $f = 0$.

f is analytic in a region containing $\mathbb{R}$, so this follows from Theorem 4.8. In particular f and $\hat{f}$ cannot both have compact support unless $f = 0$.

Theorem 3.3 (Paley–Wiener)

Suppose f is continuous and of moderate decrease on $\mathbb{R}$. Then, f has an extension to the complex plane that is entire with $|f(z)| \le Ae^{2\pi M|z|}$ for some $A > 0$, if and only if $\hat{f}$ is supported in the interval $[-M, M]$.

Proof. *(⇐)* If $\hat{f}$ is supported in $[-M, M]$, both f and $\hat{f}$ have moderate decrease and $f(x) = \int_{-M}^{M} \hat{f}(\xi)e^{2\pi i\xi x}\,d\xi$. Define $g(z) = \int_{-M}^{M} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$; $g = f$ on $\mathbb{R}$, g is entire by Theorem 5.4, and for $z = x + iy$, $|g(z)| \le \int_{-M}^{M} |\hat{f}(\xi)|e^{-2\pi\xi y}\,d\xi \le Ae^{2\pi M|z|}$.

(⇒) If $\hat{f}$ were supported in $[-M, M]$, the argument above would give the stronger bound $Ae^{2\pi M|y|}$. We reduce to that situation, adding also decay in x, and then remove the extra assumptions step by step.

Step 1. Assume f is entire with

$$|f(x + iy)| \le A\frac{e^{2\pi M|y|}}{1 + x^2}. \tag{6}$$

Then $\hat{f}(\xi) = 0$ for $|\xi| > M$. For $\xi > M$, shifting the real line down by $y > 0$ (as in (1)),

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - iy)e^{-2\pi i(x-iy)\xi}\,dx, \qquad |\hat{f}(\xi)| \le \int_{-\infty}^{\infty} A\frac{e^{2\pi My - 2\pi y\xi}}{1 + x^2}\,dx \le Ce^{-2\pi y(\xi - M)},$$

and letting $y \to \infty$ gives $\hat{f}(\xi) = 0$. For $\xi < -M$ shift up.

Step 2. Assume only

$$|f(x + iy)| \le Ae^{2\pi M|y|}. \tag{7}$$

Let $\xi > M$ and for $\epsilon > 0$ set $f_\epsilon(z) = \dfrac{f(z)}{(1 + i\epsilon z)^2}$. The factor $1/(1 + i\epsilon z)^2$ has absolute value ≤ 1 in the closed lower half-plane and tends to 1 as $\epsilon \to 0$; hence $\hat{f}_\epsilon(\xi) \to \hat{f}(\xi)$, since $|\hat{f}_\epsilon(\xi) - \hat{f}(\xi)| \le \int |f(x)|\left|\frac{1}{(1+i\epsilon x)^2} - 1\right| dx \to 0$ by moderate decrease. Each f_ϵ satisfies $|f_\epsilon(x + iy)| \le A'\frac{e^{2\pi M|y|}}{1+x^2}$ in the lower half-plane, which is where Step 1 shifts the contour for $\xi > M$; so $\hat{f}_\epsilon(\xi) = 0$, and hence $\hat{f}(\xi) = 0$. For $\xi < -M$ argue in the upper half-plane with $1/(1 - i\epsilon z)^2$.

Step 3. It remains to show the hypotheses imply (7). Dividing by a constant, it suffices: if $|f(x)| \le 1$ for real x and $|f(z)| \le e^{2\pi M|z|}$ for all z, then $|f(x+iy)| \le e^{2\pi M|y|}$. This follows from the Phragmén–Lindelöf theorem below, rotated to the first quadrant $Q = \{x > 0, y > 0\}$: $F(z) = f(z)e^{2\pi iMz}$ is

bounded by 1 on the positive real and imaginary axes and $|F(z)| \le Ce^{c|z|}$ in Q, so $|F(z)| \le 1$ in Q, i.e. $|f(z)| \le e^{2\pi My}$. The other quadrants are similar. ■

The idea of Phragmén and Lindelöf adapts the maximum modulus principle to unbounded regions:

Theorem 3.4 (Phragmén–Lindelöf)

Suppose F is a holomorphic function in the sector $S = \{z : -\pi/4 < \arg z < \pi/4\}$ that is continuous on the closure of S. Assume $|F(z)| \le 1$ on the boundary of the sector, and that there are constants $C, c > 0$ such that $|F(z)| \le Ce^{c|z|}$ for all z in the sector. Then $|F(z)| \le 1$ for all $z \in S$.

Some growth restriction is needed: $F(z) = e^{z^2}$ is bounded by 1 on ∂S but unbounded on $\mathbb{R}_+$.

Proof. Subdue the "enemy" e^{z^2}: replace it by e^{z^α} with $\alpha < 2$; take $\alpha = 3/2$. For $\epsilon > 0$ let $F_\epsilon(z) = F(z)e^{-\epsilon z^{3/2}}$, with the principal branch, so $z^{3/2} = r^{3/2}e^{3i\theta/2}$ for $z = re^{i\theta}$, $-\pi < \theta < \pi$. F_ϵ is holomorphic in S and continuous up to the boundary, and $|e^{-\epsilon z^{3/2}}| = e^{-\epsilon r^{3/2}\cos(3\theta/2)}$. In the sector $-\pi/4 < \theta < \pi/4$ we have $-\frac{\pi}{2} < -\frac{3\pi}{8} < \frac{3\theta}{2} < \frac{3\pi}{8} < \frac{\pi}{2}$, so $\cos(3\theta/2)$ is bounded below by a positive constant. With $|F(z)| \le Ce^{c|z|}$, F_ϵ decreases rapidly as $|z| \to \infty$ in the closed sector and is bounded. Let $M = \sup_{z\in\overline{S}}|F_\epsilon(z)|$; if $F \not\equiv 0$, take w_j with $|F_\epsilon(w_j)| \to M$. Since $M \neq 0$ and F_ϵ decays at infinity, the w_j accumulate at some $w \in \overline{S}$. By the maximum principle $w \notin S$, so $w \in \partial S$, where $|F| \le 1$ and $|e^{-\epsilon z^{3/2}}| \le 1$; so $M \le 1$. Thus $|F_\epsilon| \le 1$ on $\overline{S}$, and letting $\epsilon \to 0$ gives $|F| \le 1$. ■

Another version of the Paley–Wiener idea characterizes Fourier transforms vanishing for $\xi < 0$:

Theorem 3.5

Suppose f and $\hat{f}$ have moderate decrease. Then $\hat{f}(\xi) = 0$ for all $\xi < 0$ if and only if f can be extended to a continuous and bounded function in the closed upper half-plane $\{z = x + iy : y \ge 0\}$ with f holomorphic in the interior.

Proof. If $\hat{f}(\xi) = 0$ for $\xi < 0$, then $f(x) = \int_0^\infty \hat{f}(\xi)e^{2\pi i\xi x}\,d\xi$, and we extend f for $y \ge 0$ by $f(z) = \int_0^\infty \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$. The integral converges, and $|f(z)| \le A\int_0^\infty \frac{d\xi}{1+\xi^2} < \infty$, so f is bounded. The functions $f_n(z) = \int_0^n \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$ converge to f uniformly in the closed half-plane, which gives continuity there and holomorphy in the interior.

Conversely, in the spirit of Theorem 3.3, for $\epsilon, \delta > 0$ set $f_{\epsilon,\delta}(z) = \dfrac{f(z + i\delta)}{(1 - i\epsilon z)^2}$, holomorphic in a region containing the closed upper half-plane. As before, by Cauchy's theorem (shifting the contour up), $\hat{f}_{\epsilon,\delta}(\xi) = 0$ for all $\xi < 0$. Passing to the limit successively, $\hat{f}_{\epsilon,0}(\xi) = 0$ for $\xi < 0$, and finally $\hat{f}(\xi) = \hat{f}_{0,0}(\xi) = 0$ for $\xi < 0$. ■

Remark

Compare Theorem 7.1: there, for a function holomorphic in a disc, the Fourier coefficients vanish for $n < 0$; here, for a function holomorphic in the upper half-plane, the Fourier transform vanishes for $\xi < 0$.

Chapter 5. Entire Functions

Overview

Entire functions are holomorphic in the whole complex plane. The chapter is organized around three questions.

1. *Where can they vanish?* The obvious necessary condition is sufficient: for any sequence $\{z_n\}$ with no limit point in $\mathbb{C}$ there is an entire function vanishing exactly there. The construction is inspired by Euler's product for $\sin \pi z$, refined by the *Weierstrass canonical factors*.
2. *How do they grow at infinity?* The larger a function is, the more zeros it can have; for a polynomial of degree d, $\sup_{|z|=R}|P(z)| \approx R^d$ and P has d zeros. A precise version is *Jensen's formula*, linking the number of zeros in a disc to the logarithmic average of $|f|$ on the circle.
3. *To what extent are they determined by their zeros?* If f has finite order, it is determined by its zeros up to a simple factor: the *Hadamard factorization theorem*.

1 Jensen's formula

Motivation. How many zeros can an entire function have in a disc of radius R? Jensen's formula relates the zeros of f to the average of $\log|f|$ on the boundary circle, so growth controls the number of zeros.

D_R and C_R denote the open disc and circle of radius R centered at 0. We exclude the function that vanishes identically.

Theorem 1.1 (Jensen's formula)

Let Ω be an open set that contains the closure of a disc D_R and suppose that f is holomorphic in Ω, $f(0) \neq 0$, and f vanishes nowhere on the circle C_R. If $z_1, \dots, z_N$ denote the zeros of f inside the disc (counted with multiplicities), then

$$\log|f(0)| = \sum_{k=1}^{N} \log\left(\frac{|z_k|}{R}\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta. \tag{1}$$

Proof.

Step 1. If f_1, f_2 satisfy the hypotheses and (1), so does $f_1 f_2$, since $\log xy = \log x + \log y$ for $x, y > 0$ and the zeros of $f_1 f_2$ are the union of those of f_1 and f_2.

Step 2. $g(z) = f(z)/((z-z_1)\cdots(z-z_N))$, defined on $\Omega - \{z_1, \dots, z_N\}$, is bounded near each z_j, so each z_j is a removable singularity: $f(z) = (z-z_1)\cdots(z-z_N)g(z)$ with g holomorphic in Ω and nowhere vanishing on $\overline{D_R}$. By Step 1 it suffices to prove (1) for such g and for functions $z - z_j$.

Step 3. For g nowhere vanishing on $\overline{D_R}$ we must show $\log|g(0)| = \frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta$. In a slightly larger disc, $g = e^h$ with h holomorphic (discs are simply connected; Theorem 6.2). Then $|g| = e^{\mathrm{Re}(h)}$, so $\log|g| = \mathrm{Re}(h)$, and the mean value property (Corollary 7.3) gives the formula.

Step 4. For $f(z) = z - w$, $w \in D_R$, we must show $\log|w| = \log\frac{|w|}{R} + \frac{1}{2\pi}\int_0^{2\pi} \log|Re^{i\theta} - w|\, d\theta$. Since $\log(|w|/R) = \log|w| - \log R$ and $\log|Re^{i\theta} - w| = \log R + \log|e^{i\theta} - w/R|$, it suffices to show $\int_0^{2\pi} \log|e^{i\theta} - a|\, d\theta = 0$ for $|a| < 1$, equivalently (via $\theta \mapsto -\theta$, $|e^{i\theta} - a| = |1 - ae^{-i\theta}|$) $\int_0^{2\pi} \log|1 - ae^{i\theta}|\, d\theta = 0$. $F(z) = 1 - az$ vanishes nowhere on the closed unit disc, so $F = e^G$ in a disc of radius > 1 and $\log|F| = \mathrm{Re}(G)$ is harmonic. Since $\log|F(0)| = 0$, the mean value property gives the result. ∎

For f holomorphic on $\overline{D_R}$, let $n(r) = n_f(r)$ be the number of zeros of f (with multiplicities) in D_r, $0 < r < R$; $n(r)$ is non-decreasing. If $f(0) \neq 0$ and f does not vanish on C_R, then

$$\int_0^R n(r)\frac{dr}{r} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\,d\theta - \log|f(0)|, \tag{2}$$

which follows from Jensen's formula and:

Lemma 1.2

If $z_1, \ldots, z_N$ are the zeros of f inside the disc D_R, then

$$\int_0^R n(r)\frac{dr}{r} = \sum_{k=1}^{N} \log\left|\frac{R}{z_k}\right|.$$

Proof. $\sum_{k=1}^N \log\left|\frac{R}{z_k}\right| = \sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r}$. With $\eta_k(r) = 1$ if $r > |z_k|$ and 0 otherwise, $\sum_k \eta_k(r) = n(r)$, so $\sum_k \int_{|z_k|}^R \frac{dr}{r} = \sum_k \int_0^R \eta_k(r)\frac{dr}{r} = \int_0^R n(r)\frac{dr}{r}$. ∎

2 Functions of finite order

Motivation. Measuring growth by an order ρ with $|f(z)| \leq Ae^{B|z|^\rho}$, Jensen's formula gives a quantitative bound on the number of zeros. This prepares the factorization theorems of the chapter.

Definition (order of growth)

An entire function f has an **order of growth** $\leq \rho$ if $|f(z)| \leq Ae^{B|z|^\rho}$ for all $z \in \mathbb{C}$, for some constants $A, B > 0$. The **order of growth** of f is $\rho_f = \inf \rho$ over all such $\rho > 0$.

For example, e^{z^2} has order of growth 2.

Theorem 2.1

If f is an entire function that has an order of growth $\leq \rho$, then:
(i) $n(r) \leq Cr^\rho$ for some $C > 0$ and all sufficiently large r.
(ii) If $z_1, z_2, \ldots$ denote the zeros of f, with $z_k \neq 0$, then for all $s > \rho$ we have $\displaystyle\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty$.

Proof. It suffices to treat $f(0) \neq 0$: otherwise use $F(z) = f(z)/z^\ell$, ℓ the order of the zero at 0; n_f and n_F differ by a constant and F also has order of growth $\leq \rho$. By (2) with $R = 2r$,

$$\int_r^{2r} n(x)\frac{dx}{x} \leq \int_0^{2r} n(x)\frac{dx}{x} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(2re^{i\theta})|\,d\theta - \log|f(0)|.$$

Since n is increasing, the left side is $\geq n(r)\int_r^{2r}\frac{dx}{x} = n(r)\log 2$; and by the growth condition $\int_0^{2\pi}\log|f(2re^{i\theta})|\,d\theta \leq \int_0^{2\pi}\log|Ae^{B(2r)^\rho}|\,d\theta \leq C'r^\rho$ for large r. Hence $n(r) \leq Cr^\rho$. For (ii),

$$\sum_{|z_k|\geq 1} |z_k|^{-s} = \sum_{j=0}^{\infty} \sum_{2^j \leq |z_k| < 2^{j+1}} |z_k|^{-s} \leq \sum_{j=0}^{\infty} 2^{-js} n(2^{j+1}) \leq c\sum_{j=0}^{\infty} 2^{-js}2^{(j+1)\rho} \leq c'\sum_{j=0}^{\infty}(2^{\rho-s})^j < \infty,$$

since $s > \rho$. ∎

The condition $s > \rho$ cannot be improved:

Example 1

$f(z) = \sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i}$ satisfies $|f(z)| \le e^{\pi|z|}$, and taking $z = ix$ shows the order of growth is exactly 1. f vanishes to order 1 at each $n \in \mathbb{Z}$, and $\sum_{n\neq 0} 1/|n|^s < \infty$ precisely when $s > 1$.

Example 2

$f(z) = \cos z^{1/2} = \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{(2n)!}$ is entire with $|f(z)| \le e^{|z|^{1/2}}$, of order $1/2$. It vanishes at $z_n = ((n + 1/2)\pi)^2$, and $\sum_n 1/|z_n|^s < \infty$ exactly when $s > 1/2$.

Given $z_1, z_2, \ldots$, is there an entire f with zeros exactly there? Necessarily $\lim_{k\to\infty} |z_k| = \infty$ (Theorem 4.8). Weierstrass showed this suffices, by inserting factors into $(z-z_1)(z-z_2)\cdots$ that guarantee convergence without introducing new zeros.

3 Infinite products

Motivation. A polynomial is determined by its zeros up to a constant, through $\prod(z - a_k)$. To do the same for entire functions with infinitely many zeros we need infinite products and conditions for their convergence.

3.1 Generalities

For complex a_n, the product $\prod_{n=1}^{\infty}(1 + a_n)$ **converges** if $\lim_{N\to\infty} \prod_{n=1}^{N}(1 + a_n)$ exists.

Proposition 3.1

If $\sum |a_n| < \infty$, then the product $\prod_{n=1}^{\infty}(1 + a_n)$ converges. Moreover, the product converges to 0 if and only if one of its factors is 0.

Proof. For large n, $|a_n| < 1/2$; discarding finitely many terms, assume this for all n. Define $\log(1 + a_n)$ by the power series (6) of Chapter 3, so $1 + z = e^{\log(1+z)}$ for $|z| < 1$. Then $\prod_{n=1}^{N}(1 + a_n) = \prod_{n=1}^{N} e^{\log(1+a_n)} = e^{B_N}$ with $B_N = \sum_{n=1}^{N} b_n$, $b_n = \log(1 + a_n)$. From the power series, $|\log(1 + z)| \le 2|z|$ for $|z| < 1/2$, so $|b_n| \le 2|a_n|$ and $B_N \to B$. By continuity of the exponential, $e^{B_N} \to e^B$. If $1 + a_n \neq 0$ for all n, the limit is $e^B \neq 0$. ∎

Proposition 3.2

Suppose $\{F_n\}$ is a sequence of holomorphic functions on the open set Ω. If there exist constants $c_n > 0$ such that $\sum c_n < \infty$ and $|F_n(z) - 1| \le c_n$ for all $z \in \Omega$, then:
(i) The product $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$.
(ii) If $F_n(z)$ does not vanish for any n, then

$$\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F_n'(z)}{F_n(z)}.$$

Proof. (i) Write $F_n(z) = 1 + a_n(z)$ with $|a_n(z)| \le c_n$ and argue as in Proposition 3.1; the estimates are uniform in z since the c_n are constants. The uniform limit is holomorphic (Theorem 5.2).

(ii) Let $K \subset \Omega$ be compact and $G_N = \prod_{n=1}^{N} F_n$. Since $G_N \to F$ uniformly, Theorem 5.3 gives $G_N' \to F'$ uniformly on K. G_N is uniformly bounded below on K, so $G_N'/G_N \to F'/F$ uniformly on K; and $G_N'/G_N = \sum_{n=1}^{N} F_n'/F_n$ (Section 4). ∎

3.2 Example: the product formula for the sine function

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \Big(1 - \frac{z^2}{n^2}\Big) \tag{3}$$

for all $z \in \mathbb{C}$; it is derived from the sum formula for the cotangent

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N\to\infty} \sum_{|n|\le N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}, \tag{4}$$

valid for $z \notin \mathbb{Z}$. The halves of $\sum 1/(z+n)$ with $n > 0$ and $n < 0$ diverge separately; only the symmetric interpretation converges.

Proof of (4). $F(z) = \pi \cot \pi z$ has the properties:

(i) $F(z+1) = F(z)$ for $z \notin \mathbb{Z}$;

(ii) $F(z) = \frac{1}{z} + F_0(z)$ with F_0 analytic near 0;

(iii) F has simple poles at the integers and no other singularities. The function $\sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_N \sum_{|n|\le N} \frac{1}{z+n}$ has the same three properties: for (i),

$$\sum_{|n|\le N} \frac{1}{z+1+n} = \frac{1}{z+1+N} - \frac{1}{z-N} + \sum_{|n|\le N} \frac{1}{z+n},$$

and let $N \to \infty$; (ii), (iii) are evident from $\frac{1}{z} + \sum_{n\ge 1} \frac{2z}{z^2-n^2}$. Hence $\Delta(z) = F(z) - \sum \frac{1}{z+n}$ is periodic with period 1, has a removable singularity at 0 by (ii), hence at all integers by periodicity, and is entire. It suffices to show Δ is bounded, hence (by periodicity) bounded in the strip $|\mathrm{Re}(z)| \le 1/2$. It is bounded in the rectangle $|\mathrm{Im}(z)| \le 1$. For $z = x + iy$ with $y > 1$,

$$\pi \cot \pi z = \pi i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = \pi i \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}}$$

is bounded, and

$$\Big|\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}\Big| = \Big|\frac{1}{x+iy} + \sum_{n=1}^{\infty} \frac{2(x+iy)}{x^2 - y^2 - n^2 + 2ixy}\Big| \le C + C\sum_{n=1}^{\infty} \frac{y}{y^2+n^2}.$$

The last sum is majorized by $\int_0^{\infty} \frac{y}{y^2+x^2}\,dx$ (the integrand is decreasing in x), which is independent of y (change variables $x \mapsto yx$). Similarly for $y < -1$. So Δ is bounded on $\mathbb{C}$, hence constant by Liouville; Δ is odd, so the constant is 0. ∎

Proof of (3). Let $G(z) = \sin \pi z$ and $P(z) = \pi z \prod_{n=1}^{\infty}(1 - z^2/n^2)$. Since $\sum 1/n^2 < \infty$, Proposition 3.2 shows P converges and, away from the integers, $\frac{P'(z)}{P(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}$. Since $G'/G = \pi \cot \pi z$, (4) gives

$$\Big(\frac{P}{G}\Big)' = \frac{P}{G}\Big[\frac{P'}{P} - \frac{G'}{G}\Big] = 0,$$

so $P = cG$. Dividing by z and letting $z \to 0$, $c = 1$. ∎

4 Weierstrass infinite products

Motivation. The naive product $\prod(1 - z/a_n)$ may diverge. Weierstrass's canonical factors repair this, so that an entire function with any prescribed sequence of zeros (tending to infinity) exists.

Theorem 4.1 (Weierstrass)

Given any sequence $\{a_n\}$ of complex numbers with $|a_n| \to \infty$ as $n \to \infty$, there exists an entire function f that vanishes at all $z = a_n$ and nowhere else. Any other such entire function is of the form $f(z)e^{g(z)}$, where g is entire.

Repetitions are allowed in $\{a_n\}$, so this gives entire functions with prescribed zeros *and multiplicities* (the multiplicity of a zero a is the m with $f(z) = (z-a)^m g(z)$, $g(a) \neq 0$).

Proof. If f_1, f_2 both vanish exactly at the a_n (with multiplicities), f_1/f_2 has removable singularities there, so it is entire and nowhere vanishing; hence $f_1/f_2 = e^g$ for an entire g (Theorem 6.2). This proves the last statement.

To construct f: the naive product $\prod(1 - z/a_n)$ converges only for suitable $\{a_n\}$, so we insert exponential factors. Define the **canonical factors** of degree k:

$$E_0(z) = 1 - z, \qquad E_k(z) = (1-z)e^{z+z^2/2+\cdots+z^k/k} \quad (k \geq 1).$$

Suppose f is to have a zero of order m at the origin and $a_1, a_2, \ldots$ are non-zero. The **Weierstrass product** is

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Fix $R > 0$ and $|z| < R$. Factors with $|a_n| \leq 2R$ are finitely many, and their product vanishes exactly at the a_n with $|a_n| < R$. If $|a_n| > 2R$ then $|z/a_n| \leq 1/2$, and by Lemma 4.2 (with c independent of n)

$$|1 - E_n(z/a_n)| \leq c\Big|\frac{z}{a_n}\Big|^{n+1} \leq \frac{c}{2^{n+1}}.$$

By Section 3, $\prod_{|a_n|>2R} E_n(z/a_n)$ is holomorphic and non-vanishing in $|z| < R$. As R is arbitrary, f has the required properties. ∎

Lemma 4.2

If $|z| \leq 1/2$, then $|1 - E_k(z)| \leq c|z|^{k+1}$ for some $c > 0$.

Proof. For $|z| \leq 1/2$, $1 - z = e^{\log(1-z)}$ (power series), so $E_k(z) = e^{\log(1-z)+z+z^2/2+\cdots+z^k/k} = e^w$ with $w = -\sum_{n=k+1}^{\infty} z^n/n$. Since $|z| \leq 1/2$,

$$|w| \leq |z|^{k+1} \sum_{n=k+1}^{\infty} |z|^{n-k-1}/n \leq |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} \leq 2|z|^{k+1}.$$

In particular $|w| \leq 1$, so $|1 - E_k(z)| = |1 - e^w| \leq c'|w| \leq c|z|^{k+1}$. ∎

Remark

The constant c is independent of k: one may take $c' = e$, $c = 2e$.

5 Hadamard's factorization theorem

Weierstrass's theorem writes a function with zeros $a_1, a_2, \ldots$ as $e^{g(z)} z^m \prod E_n(z/a_n)$. Hadamard showed that for functions of finite order the degree of the canonical factors can be taken constant, and g is then a polynomial. Recall that if f has order of growth $\leq \rho$, then $n(r) \leq Cr^\rho$ for large r and $\sum |a_n|^{-s} < \infty$ for $s > \rho$ (Theorem 2.1).

Theorem 5.1 (Hadamard)

Suppose f is entire and has growth order ρ_0. Let k be the integer so that $k \le \rho_0 < k+1$. If $a_1, a_2, \dots$ denote the (non-zero) zeros of f, then

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\le k$, and m is the order of the zero of f at $z = 0$.

Main lemmas.

Lemma 5.2

The canonical products satisfy

$$|E_k(z)| \ge e^{-c|z|^{k+1}} \text{ if } |z| \le 1/2 \qquad \text{and} \qquad |E_k(z)| \ge |1-z| e^{-c'|z|^k} \text{ if } |z| \ge 1/2.$$

Proof. If $|z| \le 1/2$, $E_k(z) = e^{\log(1-z) + \sum_{n=1}^{k} z^n/n} = e^{-\sum_{n=k+1}^{\infty} z^n/n} = e^w$, and $|e^w| \ge e^{-|w|}$ with $|w| \le c|z|^{k+1}$. If $|z| \ge 1/2$, $|E_k(z)| = |1-z||e^{z+z^2/2+\cdots+z^k/k}|$, and $|e^{z+\cdots+z^k/k}| \ge e^{-|z+\cdots+z^k/k|} \ge e^{-c'|z|^k}$ (as $|z| \ge 1/2$). ∎

The key is a lower bound for the product of canonical factors away from the zeros.

Lemma 5.3

For any s with $\rho_0 < s < k+1$, we have

$$\Big| \prod_{n=1}^{\infty} E_k(z/a_n) \Big| \ge e^{-c|z|^s},$$

except possibly when z belongs to the union of the discs centered at a_n of radius $|a_n|^{-k-1}$, for $n = 1, 2, 3, \dots$.

Proof. Write $\prod_n E_k(z/a_n) = \prod_{|a_n| \le 2|z|} E_k(z/a_n) \prod_{|a_n| > 2|z|} E_k(z/a_n)$. For the second product, with no restriction on z, Lemma 5.2 gives

$$\Big| \prod_{|a_n|>2|z|} E_k(z/a_n) \Big| \ge \prod_{|a_n|>2|z|} e^{-c|z/a_n|^{k+1}} = e^{-c|z|^{k+1} \sum_{|a_n|>2|z|} |a_n|^{-k-1}},$$

and since $|a_n| > 2|z|$ and $s < k+1$, $|a_n|^{-k-1} = |a_n|^{-s}|a_n|^{s-k-1} \le C|a_n|^{-s}|z|^{s-k-1}$; as $\sum |a_n|^{-s} < \infty$, this product is $\ge e^{-c|z|^s}$. For the first product, by the second part of Lemma 5.2,

$$\Big| \prod_{|a_n| \le 2|z|} E_k(z/a_n) \Big| \ge \prod_{|a_n| \le 2|z|} \Big|1 - \frac{z}{a_n}\Big| \prod_{|a_n| \le 2|z|} e^{-c'|z/a_n|^k}. \tag{5}$$

Here $\prod_{|a_n| \le 2|z|} e^{-c'|z/a_n|^k} = e^{-c'|z|^k \sum_{|a_n| \le 2|z|} |a_n|^{-k}}$ and $|a_n|^{-k} = |a_n|^{-s}|a_n|^{s-k} \le C|a_n|^{-s}|z|^{s-k}$, so this is $\ge e^{-c|z|^s}$. For the first factor of (5) we need the restriction on z: if z is not in the disc of radius $|a_n|^{-k-1}$ about a_n, then $|a_n - z| \ge |a_n|^{-k-1}$, so

$$\prod_{|a_n| \le 2|z|} \Big|1 - \frac{z}{a_n}\Big| = \prod_{|a_n| \le 2|z|} \Big|\frac{a_n - z}{a_n}\Big| \ge \prod_{|a_n| \le 2|z|} |a_n|^{-k-1}|a_n|^{-1} = \prod_{|a_n| \le 2|z|} |a_n|^{-k-2}.$$

Finally $\log \prod_{|a_n| \le 2|z|} |a_n|^{k+2} = (k+2) \sum_{|a_n| \le 2|z|} \log|a_n| \le (k+2) n(2|z|) \log 2|z| \le c|z|^s \log 2|z| \le c'|z|^{s'}$ for any $s' > s$, using $n(2|z|) \le c|z|^s$ (Theorem 2.1). Since $s > \rho_0$ was arbitrary, starting with s close enough to ρ_0 gives the lemma (with s' in place of s). ∎

Corollary 5.4

There exists a sequence of radii $r_1, r_2, \dots$ with $r_m \to \infty$, such that

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \geq e^{-c|z|^s} \qquad \text{for } |z| = r_m.$$

Proof. Since $\sum |a_n|^{-k-1} < \infty$, there is N with $\sum_{n\geq N} |a_n|^{-k-1} < 1/10$. Given consecutive large integers L, $L+1$, there is $r \in [L, L+1]$ such that the circle of radius r about 0 meets none of the forbidden discs of Lemma 5.3: otherwise the intervals $I_n = [|a_n| - |a_n|^{-k-1}, |a_n| + |a_n|^{-k-1}]$, of length $2|a_n|^{-k-1}$, would cover $[L, L+1]$, giving $2\sum_{n\geq N} |a_n|^{-k-1} \geq 1$, a contradiction. Apply Lemma 5.3 with $|z| = r$. ∎

Proof of Hadamard's theorem. Let $E(z) = z^m \prod_{n=1}^{\infty} E_k(z/a_n)$. E is entire, as in the proof of Theorem 4.1: by Lemma 4.2, $|1 - E_k(z/a_n)| \leq c|z/a_n|^{k+1}$ for large n, and $\sum |a_n|^{-k-1}$ converges ($\rho_0 < s < k+1$). E has the zeros of f, so f/E is entire and nowhere vanishing: $f(z)/E(z) = e^{g(z)}$ for an entire g. Since f has growth order ρ_0 and by Corollary 5.4,

$$e^{\operatorname{Re}(g(z))} = \Big|\frac{f(z)}{E(z)}\Big| \leq c' e^{c|z|^s} \qquad \text{for } |z| = r_m,$$

so $\operatorname{Re}(g(z)) \leq C|z|^s$ for $|z| = r_m$. Lemma 5.5 gives that g is a polynomial of degree $\leq s$, hence $\leq k$ (as $s < k+1$). ∎

Lemma 5.5

Suppose g is entire and $u = \operatorname{Re}(g)$ satisfies $u(z) \leq Cr^s$ whenever $|z| = r$, for a sequence of positive real numbers r that tends to infinity. Then g is a polynomial of degree $\leq s$.

Proof. Write $g(z) = \sum_{n=0}^{\infty} a_n z^n$. By Theorem 7.1,

$$\frac{1}{2\pi}\int_0^{2\pi} g(re^{i\theta}) e^{-in\theta}\, d\theta = \begin{cases} a_n r^n & n \geq 0, \\ 0 & n < 0. \end{cases} \tag{6}$$

Taking complex conjugates, $\frac{1}{2\pi}\int_0^{2\pi} \overline{g(re^{i\theta})} e^{-in\theta}\, d\theta = 0$ for $n > 0$ (7). Since $2u = g + \overline{g}$, adding (6) and (7),

$$a_n r^n = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta}) e^{-in\theta}\, d\theta \quad (n > 0), \qquad 2\operatorname{Re}(a_0) = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})\, d\theta.$$

[Supplement]Since $\int_0^{2\pi} e^{-in\theta} d\theta = 0$ for $n > 0$, $a_n r^n = -\frac{1}{\pi}\int_0^{2\pi} [Cr^s - u(re^{i\theta})] e^{-in\theta}\, d\theta$, and $Cr^s - u \geq 0$ on the circles in question, so

$$|a_n| r^n \leq \frac{1}{\pi}\int_0^{2\pi} [Cr^s - u(re^{i\theta})]\, d\theta = 2Cr^s - 2\operatorname{Re}(a_0).$$

Letting $r \to \infty$ along the sequence, $a_n = 0$ for $n > s$. ∎

Chapter 6. The Gamma and Zeta Functions

Overview

The gamma and zeta functions are among the most important non-elementary functions. The gamma function is ubiquitous; its basic structural property essentially characterizes it: $1/\Gamma(s)$ is the (simplest) entire function with zeros exactly at $s = 0, -1, -2, \ldots$. The zeta function is fundamental in analytic number theory, through Euler's identity

$$\prod_p \frac{1}{1-p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s)$$

over all primes. ζ is easily analytic for $\mathrm{Re}(s) > 1$; Riemann realized that the study of primes is bound up with its meromorphic continuation to $\mathbb{C}$. We prove this and the functional equation (a symmetry about $\mathrm{Re}(s) = 1/2$, based on the theta function identity), and estimate ζ near $\mathrm{Re}(s) = 1$ for the prime number theorem (Chapter 7). Following convention, the variable is s.

1 The gamma function

Motivation. The gamma function interpolates the factorial and is the main special function of analytic number theory. Its analytic continuation, its product formula and the identity $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ are the tools needed for the zeta function.

For $s > 0$ the **gamma function** is

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1}\, dt. \tag{1}$$

The integral converges: t^{s-1} is integrable near 0, and e^{-t} decays at infinity.

Proposition 1.1

The gamma function extends to an analytic function in the half-plane $\mathrm{Re}(s) > 0$, and is still given there by the integral formula (1).

Proof. It suffices to show the integral is holomorphic in each strip $S_{\delta,M} = \{\delta < \mathrm{Re}(s) < M\}$, $0 < \delta < M < \infty$. With $\sigma = \mathrm{Re}(s)$, $|e^{-t}t^{s-1}| = e^{-t}t^{\sigma-1}$, so $\Gamma(s) = \lim_{\epsilon\to 0} \int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\, dt$ converges for $s \in S_{\delta,M}$. Let $F_\epsilon(s) = \int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\, dt$, holomorphic in $S_{\delta,M}$ by Theorem 5.4. By Theorem 5.2 it suffices that $F_\epsilon \to \Gamma$ uniformly on the strip. Now

$$|\Gamma(s) - F_\epsilon(s)| \le \int_0^\epsilon e^{-t}t^{\sigma-1}\, dt + \int_{1/\epsilon}^\infty e^{-t}t^{\sigma-1}\, dt.$$

The first integral is $\le \epsilon^\delta/\delta$ for $0 < \epsilon < 1$, and the second is $\le \int_{1/\epsilon}^\infty e^{-t}t^{M-1}\, dt \le C\int_{1/\epsilon}^\infty e^{-t/2}\, dt \to 0$, uniformly in the strip. ∎

1.1 Analytic continuation

Although the integral does not converge absolutely elsewhere, there is a meromorphic function on $\mathbb{C}$ equal to Γ for $\mathrm{Re}(s) > 0$, its **analytic continuation**, still denoted Γ (unique, since the complement of the poles of a meromorphic function is connected).

Lemma 1.2

If $\operatorname{Re}(s) > 0$, then

$$\Gamma(s+1) = s\Gamma(s). \tag{2}$$

As a consequence $\Gamma(n+1) = n!$ for $n = 0, 1, 2, \ldots$.

Proof. Integrating by parts,

$$\int_\epsilon^{1/\epsilon} \frac{d}{dt}(e^{-t}t^s)\,dt = -\int_\epsilon^{1/\epsilon} e^{-t}t^s\,dt + s\int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\,dt,$$

and letting $\epsilon \to 0$ the left side vanishes, since $e^{-t}t^s \to 0$ as $t \to 0$ or $t \to \infty$. Also $\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1$; apply (2) successively. ∎

Theorem 1.3

The function $\Gamma(s)$ initially defined for $\operatorname{Re}(s) > 0$ has an analytic continuation to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at the negative integers $s = 0, -1, \ldots$. The residue of Γ at $s = -n$ is $(-1)^n/n!$.

Proof. It suffices to extend Γ to each half-plane $\operatorname{Re}(s) > -m$, $m \geq 1$. For $\operatorname{Re}(s) > -m$ define

$$F_m(s) = \frac{\Gamma(s+m)}{(s+m-1)(s+m-2)\cdots s}.$$

F_m is meromorphic in $\operatorname{Re}(s) > -m$ with simple poles at $s = 0, -1, \ldots, -m+1$ and residues

$$\operatorname{res}_{s=-n} F_m(s) = \frac{\Gamma(-n+m)}{(m-1-n)!(-1)(-2)\cdots(-n)} = \frac{(m-n-1)!}{(m-1-n)!(-1)(-2)\cdots(-n)} = \frac{(-1)^n}{n!}.$$

By Lemma 1.2, $F_m = \Gamma$ for $\operatorname{Re}(s) > 0$, so by uniqueness $F_m = F_k$ $(1 \leq k \leq m)$ on the domain of F_k; this gives the continuation. ∎

Remark

By analytic continuation, $\Gamma(s+1) = s\Gamma(s)$ holds whenever $s \neq 0, -1, -2, \ldots$. At $s = -n$, $n \geq 1$, both sides are infinite and $\operatorname{res}_{s=-n}\Gamma(s+1) = -n\operatorname{res}_{s=-n}\Gamma(s)$. At $s = 0$, $\Gamma(1) = \lim_{s\to 0} s\Gamma(s)$.

An alternate proof. For $\operatorname{Re}(s) > 0$ split $\Gamma(s) = \int_0^1 e^{-t}t^{s-1}\,dt + \int_1^\infty e^{-t}t^{s-1}\,dt$. The second integral is entire; expanding e^{-t} and integrating termwise, $\int_0^1 e^{-t}t^{s-1}\,dt = \sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)}$, so

$$\Gamma(s) = \sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)} + \int_1^\infty e^{-t}t^{s-1}\,dt \qquad \text{for } \operatorname{Re}(s) > 0. \tag{3}$$

The series is meromorphic on $\mathbb{C}$ with poles at the negative integers and residue $(-1)^n/n!$ at $-n$: for fixed R, split it at $N > 2R$; the finite part is meromorphic in $|s| < R$ with the right poles and residues, and for $n > N$, $|s| < R$, $\left|\frac{(-1)^n}{n!(n+s)}\right| \leq \frac{1}{n!R}$, so the tail converges uniformly and is holomorphic there. Hence (3) holds on all of $\mathbb{C}$.

1.2 Further properties of Γ

Theorem 1.4

For all $s \in \mathbb{C}$,

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}. \tag{4}$$

$\Gamma(1-s)$ has simple poles at $s = 1, 2, \ldots$, so $\Gamma(s)\Gamma(1-s)$ is meromorphic with simple poles at all integers, as is $\pi/\sin \pi s$. By analytic continuation it suffices to prove (4) for $0 < s < 1$.

Lemma 1.5

For $0 < a < 1$, $\displaystyle\int_0^\infty \frac{v^{a-1}}{1+v}\, dv = \frac{\pi}{\sin \pi a}$.

Proof. With $v = e^x$, $\int_0^\infty \frac{v^{a-1}}{1+v}\, dv = \int_{-\infty}^\infty \frac{e^{ax}}{1+e^x}\, dx$, which equals $\pi/\sin \pi a$ by Example 2. ■

Proof of Theorem 1.4. For $0 < s < 1$ and $t > 0$, the change of variables $u = vt$ gives $\Gamma(1-s) = \int_0^\infty e^{-u} u^{-s}\, du = t \int_0^\infty e^{-vt}(vt)^{-s}\, dv$. Hence

$$\begin{aligned}\Gamma(1-s)\Gamma(s) &= \int_0^\infty e^{-t} t^{s-1} \Gamma(1-s)\, dt = \int_0^\infty e^{-t} t^{s-1} \Big(t \int_0^\infty e^{-vt} (vt)^{-s}\, dv \Big) dt \\ &= \int_0^\infty \int_0^\infty e^{-t[1+v]} v^{-s}\, dt\, dv = \int_0^\infty \frac{v^{-s}}{1+v}\, dv = \frac{\pi}{\sin \pi(1-s)} = \frac{\pi}{\sin \pi s}.\end{aligned}$$

■

With $s = 1/2$, since $\Gamma(s) > 0$ for $s > 0$, $\Gamma(1/2) = \sqrt{\pi}$.

Theorem 1.6

The function Γ has the following properties:
(i) $1/\Gamma(s)$ is an entire function of s with simple zeros at $s = 0, -1, -2, \ldots$ and it vanishes nowhere else.
(ii) $1/\Gamma(s)$ has growth $\left|\frac{1}{\Gamma(s)}\right| \le c_1 e^{c_2 |s| \log |s|}$. Therefore, $1/\Gamma$ is of order 1 in the sense that for every $\epsilon > 0$, there exists a bound $c(\epsilon)$ so that $\left|\frac{1}{\Gamma(s)}\right| \le c(\epsilon) e^{c_2 |s|^{1+\epsilon}}$.

Proof. By Theorem 1.4,

$$\frac{1}{\Gamma(s)} = \Gamma(1-s) \frac{\sin \pi s}{\pi}, \tag{5}$$

so the simple poles of $\Gamma(1-s)$ at $s = 1, 2, \ldots$ are cancelled by zeros of $\sin \pi s$, and $1/\Gamma$ is entire with simple zeros at $0, -1, -2, \ldots$.

For the estimate, first $\int_1^\infty e^{-t} t^\sigma\, dt \le e^{(\sigma+1)\log(\sigma+1)}$ for $\sigma > 0$: choose n with $\sigma \le n \le \sigma + 1$; then $\int_1^\infty e^{-t} t^\sigma\, dt \le \int_0^\infty e^{-t} t^n\, dt = n! \le n^n = e^{n \log n} \le e^{(\sigma+1)\log(\sigma+1)}$. Since (3) holds on $\mathbb{C}$, by (5)

$$\frac{1}{\Gamma(s)} = \Big(\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+1-s)} \Big) \frac{\sin \pi s}{\pi} + \Big(\int_1^\infty e^{-t} t^{-s}\, dt \Big) \frac{\sin \pi s}{\pi}.$$

$\left| \int_1^\infty e^{-t} t^{-s}\, dt \right| \le \int_1^\infty e^{-t} t^{|\sigma|}\, dt \le e^{(|s|+1)\log(|s|+1)}$ and $|\sin \pi s| \le e^{\pi |s|}$, so the second term is $\le c_1 e^{c_2 |s| \log |s|}$. For the first term: if $|\mathrm{Im}(s)| > 1$ it is $\le c e^{\pi |s|}$. If $|\mathrm{Im}(s)| \le 1$, let k be the integer with $k - 1/2 \le \mathrm{Re}(s) < k + 1/2$. If $k \ge 1$,

$$\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+1-s)} \frac{\sin \pi s}{\pi} = (-1)^{k-1} \frac{\sin \pi s}{\pi (k-1)!(k-s)} + \sum_{n \ne k-1} \frac{(-1)^n}{n!(n+1-s)} \frac{\sin \pi s}{\pi};$$

the first term is bounded since $\sin \pi s$ vanishes at $s = k$, and the second is majorized by $c\sum 1/n!$. If $k \le 0$ then $\mathrm{Re}(s) < 1/2$ and the series is again bounded by $c\sum 1/n!$. ∎

Theorem 1.7

For all $s \in \mathbb{C}$,

$$\frac{1}{\Gamma(s)} = e^{\gamma s} s \prod_{n=1}^{\infty} \Big(1 + \frac{s}{n}\Big) e^{-s/n}.$$

Here γ is **Euler's constant**, $\gamma = \lim_{N\to\infty} \big(\sum_{n=1}^{N} \frac{1}{n} - \log N\big)$. The limit exists: $\sum_{n=1}^{N} \frac{1}{n} - \log N = \sum_{n=1}^{N-1} \int_n^{n+1} \Big[\frac{1}{n} - \frac{1}{x}\Big] dx + \frac{1}{N}$, and by the mean value theorem $0 \le \frac{1}{n} - \frac{1}{x} \le \frac{1}{n^2}$ for $n \le x \le n+1$; so the sum is $\sum_{n=1}^{N-1} a_n + \frac{1}{N}$ with $0 \le a_n \le 1/n^2$, and $\sum a_n$ converges.

Proof. By Hadamard's theorem, since $1/\Gamma$ is entire of growth order 1 with simple zeros at $0, -1, -2, \dots$,

$$\frac{1}{\Gamma(s)} = e^{As+B} s \prod_{n=1}^{\infty} \Big(1 + \frac{s}{n}\Big) e^{-s/n}.$$

Since $s\Gamma(s) \to 1$ as $s \to 0$, $B = 0$ (or an integer multiple of $2\pi i$, which gives the same result). Putting $s = 1$ and using $\Gamma(1) = 1$,

$$e^{-A} = \prod_{n=1}^{\infty} \Big(1 + \frac{1}{n}\Big) e^{-1/n} = \lim_{N\to\infty} e^{\sum_{n=1}^{N} [\log(1+1/n) - 1/n]} = \lim_{N\to\infty} e^{-(\sum_{n=1}^{N} 1/n) + \log N + \log(1+1/N)} = e^{-\gamma}.$$

So $A = \gamma + 2\pi i k$; since $\Gamma(s)$ is real for real s, $k = 0$. ∎

Thus $1/\Gamma$ is essentially characterized (up to two normalizing constants) as the entire function with (i) simple zeros at $s = 0, -1, -2, \dots$ and no others, and (ii) order of growth 1. $\sin \pi s$ has a similar characterization (zeros at all integers), and satisfies the stricter $\sin \pi s = O(e^{c|s|})$; this estimate without the logarithm fails for $1/\Gamma$.

2 The zeta function

Motivation. The zeta function $\zeta(s) = \sum n^{-s}$ encodes the primes through the Euler product. To use complex analysis on it, we continue it to the whole plane (except a pole at $s = 1$) and prove the functional equation relating $\zeta(s)$ to $\zeta(1 - s)$.

The **Riemann zeta function** is initially defined for real $s > 1$ by $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$.

2.1 Functional equation and analytic continuation

Proposition 2.1

The series defining $\zeta(s)$ converges for $\mathrm{Re}(s) > 1$, and the function ζ is holomorphic in this half-plane.

Proof. For $s = \sigma + it$, $|n^{-s}| = |e^{-s \log n}| = e^{-\sigma \log n} = n^{-\sigma}$. If $\sigma \ge 1 + \delta > 1$ the series is uniformly bounded by $\sum 1/n^{1+\delta} < \infty$; so it converges uniformly on each half-plane $\mathrm{Re}(s) \ge 1 + \delta$ and defines a holomorphic function in $\mathrm{Re}(s) > 1$. ∎

The continuation is subtler than for Γ; it relates ζ to the **theta function** $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$, $t > 0$, which satisfies (by Poisson summation, (4) of Chapter 4) $\vartheta(t) = t^{-1/2}\vartheta(1/t)$. We need

$$\vartheta(t) \le C t^{-1/2} \quad \text{as } t \to 0, \qquad |\vartheta(t) - 1| \le C e^{-\pi t} \quad \text{for } t \ge 1 :$$

the first follows from the functional equation, the second from $\sum_{n\ge1} e^{-\pi n^2 t} \le \sum_{n\ge1} e^{-\pi n t} \le Ce^{-\pi t}$ for $t \ge 1$.

Theorem 2.2

If $\mathrm{Re}(s) > 1$, then

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}\int_0^\infty u^{(s/2)-1}[\vartheta(u) - 1]\,du.$$

Proof. The key is

$$\int_0^\infty e^{-\pi n^2 u} u^{(s/2)-1}\,du = \pi^{-s/2}\Gamma(s/2)n^{-s}, \qquad n \ge 1 : \tag{6}$$

with $u = t/\pi n^2$ the left side becomes $\left(\int_0^\infty e^{-t}t^{(s/2)-1}\,dt\right)(\pi n^2)^{-s/2}$. Since $\frac{\vartheta(u)-1}{2} = \sum_{n=1}^\infty e^{-\pi n^2 u}$, and the estimates on ϑ justify interchanging sum and integral,

$$\frac{1}{2}\int_0^\infty u^{(s/2)-1}[\vartheta(u) - 1]\,du = \sum_{n=1}^\infty \int_0^\infty u^{(s/2)-1}e^{-\pi n^2 u}\,du = \pi^{-s/2}\Gamma(s/2)\sum_{n=1}^\infty n^{-s} = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

■

The **xi function**, a more symmetric modification of ζ, is defined for $\mathrm{Re}(s) > 1$ by

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{7}$$

Theorem 2.3

The function ξ is holomorphic for $\mathrm{Re}(s) > 1$ and has an analytic continuation to all of $\mathbb{C}$ as a meromorphic function with simple poles at $s = 0$ and $s = 1$. Moreover, $\xi(s) = \xi(1 - s)$ for all $s \in \mathbb{C}$.

Proof. Heuristically: multiply $\sum_n e^{-\pi n^2 u} = u^{-1/2}\sum_n e^{-\pi n^2/u}$ by $u^{(s/2)-1}$ and integrate, using (6) and the change $u \mapsto 1/u$ (discarding the $n = 0$ terms). Precisely: let $\psi(u) = [\vartheta(u) - 1]/2$. The functional equation of ϑ gives

$$\psi(u) = u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}.$$

By Theorem 2.2, for $\mathrm{Re}(s) > 1$,

$$\begin{aligned}
\pi^{-s/2}\Gamma(s/2)\zeta(s) &= \int_0^1 u^{(s/2)-1}\psi(u)\,du + \int_1^\infty u^{(s/2)-1}\psi(u)\,du \\
&= \int_0^1 u^{(s/2)-1}\Big[u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}\Big]du + \int_1^\infty u^{(s/2)-1}\psi(u)\,du \\
&= \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1})\psi(u)\,du
\end{aligned}$$

(substituting $u \mapsto 1/u$ in the first integral). Therefore

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1})\psi(u)\,du.$$

Since ψ decays exponentially at infinity, the integral is entire, so ξ continues to $\mathbb{C}$ with simple poles at $s = 0$, $s = 1$. The integral is unchanged under $s \mapsto 1 - s$, and so is $\frac{1}{s-1} - \frac{1}{s}$; hence $\xi(s) = \xi(1-s)$.
■

Theorem 2.4

The zeta function has a meromorphic continuation into the entire complex plane, whose only singularity is a simple pole at $s = 1$.

Proof. By (7), $\zeta(s) = \pi^{s/2}\xi(s)/\Gamma(s/2)$. $1/\Gamma(s/2)$ is entire with simple zeros at $0, -2, -4, \ldots$, so the simple pole of ξ at 0 is cancelled, and the only singularity of ζ is a simple pole at $s = 1$. ■

A more elementary approach gives the continuation to $\mathrm{Re}(s) > 0$ and is useful for the growth of ζ near $\mathrm{Re}(s) = 1$: compare $\sum n^{-s}$ with $\int_1^\infty x^{-s}\,dx$.

Proposition 2.5

There is a sequence of entire functions $\{\delta_n(s)\}_{n=1}^\infty$ that satisfy the estimate $|\delta_n(s)| \le |s|/n^{\sigma+1}$, where $s = \sigma + it$, and such that

$$\sum_{1\le n<N} \frac{1}{n^s} - \int_1^N \frac{dx}{x^s} = \sum_{1\le n<N} \delta_n(s), \tag{8}$$

whenever N is an integer > 1.

Proof. Set

$$\delta_n(s) = \int_n^{n+1} \Big[\frac{1}{n^s} - \frac{1}{x^s}\Big]dx. \tag{9}$$

By the mean-value theorem applied to x^{-s}, $|\frac{1}{n^s} - \frac{1}{x^s}| \le \frac{|s|}{n^{\sigma+1}}$ for $n \le x \le n+1$, so $|\delta_n(s)| \le |s|/n^{\sigma+1}$; and $\int_1^N \frac{dx}{x^s} = \sum_{1\le n<N} \int_n^{n+1} \frac{dx}{x^s}$. ■

Corollary 2.6

For $\mathrm{Re}(s) > 0$ we have

$$\zeta(s) - \frac{1}{s-1} = H(s),$$

where $H(s) = \sum_{n=1}^\infty \delta_n(s)$ is holomorphic in the half-plane $\mathrm{Re}(s) > 0$.

Proof. First let $\mathrm{Re}(s) > 1$ and $N \to \infty$ in (8): $\int_1^N x^{-s}dx \to \frac{1}{s-1}$, and by $|\delta_n(s)| \le |s|/n^{\sigma+1}$ the series $\sum \delta_n(s)$ converges uniformly on compact subsets of $\mathrm{Re}(s) \ge \delta > 0$. This proves the identity for $\mathrm{Re}(s) > 1$; the uniform convergence shows H is holomorphic in $\mathrm{Re}(s) > 0$, so ζ extends there and the identity persists. ■

For $\mathrm{Re}(s) > 1$, $|\zeta(s)| \le \sum n^{-\sigma}$, so ζ is bounded in $\mathrm{Re}(s) \ge 1 + \delta$. The growth near $\mathrm{Re}(s) = 1$ is mild (the following estimates are crude but suffice for Chapter 7).

Proposition 2.7

Suppose $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$. Then for each σ_0, $0 \le \sigma_0 \le 1$, and every $\epsilon > 0$, there exists a constant c_ϵ so that (i) $|\zeta(s)| \le c_\epsilon |t|^{1-\sigma_0+\epsilon}$, if $\sigma_0 \le \sigma$ and $|t| \ge 1$.
(ii) $|\zeta'(s)| \le c_\epsilon |t|^\epsilon$, if $1 \le \sigma$ and $|t| \ge 1$.

In particular $\zeta(1 + it) = O(|t|^\epsilon)$ as $|t| \to \infty$, and likewise for ζ'.

Proof. Besides $|\delta_n(s)| \le |s|/n^{\sigma+1}$ we have $|\delta_n(s)| \le 2/n^\sigma$, from (9) and $|n^{-s}| = n^{-\sigma}$, $|x^{-s}| \le n^{-\sigma}$ for $x \ge n$. Combining via $A = A^\delta A^{1-\delta}$ $(0 \le \delta \le 1)$,

$$|\delta_n(s)| \le \Big(\frac{|s|}{n^{\sigma_0+1}}\Big)^\delta \Big(\frac{2}{n^{\sigma_0}}\Big)^{1-\delta} \le \frac{2|s|^\delta}{n^{\sigma_0+\delta}} \qquad (\sigma \ge \sigma_0).$$

Take $\delta = 1 - \sigma_0 + \epsilon$ and use Corollary 2.6: for $\sigma \geq \sigma_0$,

$$|\zeta(s)| \leq \Big|\frac{1}{s-1}\Big| + 2|s|^{1-\sigma_0+\epsilon} \sum_{n=1}^{\infty} \frac{1}{n^{1+\epsilon}},$$

which gives (i). For (ii), by the Cauchy integral formula on the circle of radius r about s,

$$\zeta'(s) = \frac{1}{2\pi r} \int_0^{2\pi} \zeta(s + re^{i\theta}) e^{-i\theta}\, d\theta.$$

Take $r = \epsilon$; this circle lies in $\mathrm{Re} \geq 1 - \epsilon$, and (ii) follows from (i) (with $\sigma_0 = 1 - \epsilon$ and 2ϵ in place of ϵ). ■

Chapter 7. The Zeta Function and Prime Number Theorem

Overview

Euler's product formula for ζ connects analysis with the arithmetic of primes; an easy consequence is that $\sum_p 1/p$ diverges. To understand the distribution of primes, let

$$\pi(x) = \text{number of primes less than or equal to } x.$$

Legendre and Gauss conjectured

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as } x \to \infty, \tag{1}$$

where $f(x) \sim g(x)$ means $f(x)/g(x) \to 1$. Tchebychev proved by elementary methods the weaker $\pi(x) \approx x/\log x$, i.e. $A\frac{x}{\log x} \le \pi(x) \le B\frac{x}{\log x}$ for large x (2). In 1896 Hadamard and de la Vallée Poussin proved (1), the *prime number theorem*, using complex analysis, as we do. At the heart of the proof lies the fact that $\zeta(s)$ does not vanish on the line $\mathrm{Re}(s) = 1$ (the two statements are in fact equivalent).

1 Zeros of the zeta function

Motivation. The prime number theorem $\pi(x) \sim x/\log x$ is equivalent to the fact that ζ has no zeros on the line $\mathrm{Re}\, s = 1$. We prove this nonvanishing and the estimates for $1/\zeta$ needed to carry out the contour integration.

Euler's identity. For $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Proof. Each factor is a convergent geometric series $1 + p^{-s} + p^{-2s} + \cdots$; formally multiplying them gives the result. Precisely, take first $s > 1$ real and integers $M > N$. By unique factorization, every $n \le N$ is a product of primes $\le N$, each repeated fewer than M times, so

$$\sum_{n=1}^{N} \frac{1}{n^s} \le \prod_{p \le N} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots + \frac{1}{p^{Ms}}\right) \le \prod_{p \le N} \frac{1}{1 - p^{-s}} \le \prod_p \frac{1}{1 - p^{-s}},$$

and letting $N \to \infty$, $\sum_{n=1}^{\infty} n^{-s} \le \prod_p (1 - p^{-s})^{-1}$. Conversely, again by unique factorization, $\prod_{p \le N}(1 + p^{-s} + \cdots + p^{-Ms}) \le \sum_{n=1}^{\infty} n^{-s}$; letting $M \to \infty$, then $N \to \infty$, gives the reverse inequality. (For complex s with $\mathrm{Re}(s) > 1$ the identity follows by analytic continuation, or by the same argument with absolute convergence.) ∎

By Proposition 3.1, $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$. The functional equation $\xi(s) = \xi(1 - s)$ reads $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta(1-s)$, so

$$\zeta(s) = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}\zeta(1-s).$$

For $\mathrm{Re}(s) < 0$:

(i) $\zeta(1 - s)$ has no zeros, as $\mathrm{Re}(1 - s) > 1$;

(ii) $\Gamma((1 - s)/2)$ is zero free;

(iii) $1/\Gamma(s/2)$ has zeros at $s = -2, -4, -6, \ldots$. Hence:

Theorem 1.1

The only zeros of ζ outside the strip $0 \le \operatorname{Re}(s) \le 1$ are at the negative even integers, $-2, -4, -6, \dots$.

The region $0 \le \operatorname{Re}(s) \le 1$ is the **critical strip**; the zeros outside it are the **trivial zeros**. Riemann conjectured the **Riemann hypothesis**: *the zeros of $\zeta(s)$ in the critical strip lie on the line* $\operatorname{Re}(s) = 1/2$—still one of the most famous unsolved problems. We prove:

Theorem 1.2

The zeta function has no zeros on the line $\operatorname{Re}(s) = 1$.

(Since ζ has a pole at 1 there are no zeros near $s = 1$; the point is the deeper fact $\zeta(1 + it) \neq 0$ for all $t \in \mathbb{R}$.) By the functional equation, ζ then has no zeros on $\operatorname{Re}(s) = 0$ either.

Lemma 1.3

If $\operatorname{Re}(s) > 1$, then

$$\log \zeta(s) = \sum_{p,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n n^{-s}$$

for some $c_n \ge 0$.

Proof. For $s > 1$, take the logarithm of Euler's product and use $\log \frac{1}{1-x} = \sum_{m=1}^{\infty} \frac{x^m}{m}$ $(0 \le x < 1)$:

$$\log \zeta(s) = \sum_{p} \log\Big(\frac{1}{1 - p^{-s}}\Big) = \sum_{p,m} \frac{p^{-ms}}{m}.$$

The double sum converges absolutely, so the order of summation is irrelevant (see the Note at the end of the chapter). The formula holds for $\operatorname{Re}(s) > 1$ by analytic continuation; $\log \zeta$ is well defined in the simply connected half-plane $\operatorname{Re}(s) > 1$ since ζ has no zeros there (Theorem 6.2). Finally $c_n = 1/m$ if $n = p^m$ and $c_n = 0$ otherwise. ∎

Lemma 1.4

If $\theta \in \mathbb{R}$, then $3 + 4\cos\theta + \cos 2\theta \ge 0$.

This follows from $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2$.

Corollary 1.5

If $\sigma > 1$ and t is real, then $\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \ge 0$.

Proof. With $s = \sigma + it$, $\operatorname{Re}(n^{-s}) = \operatorname{Re}(e^{-(\sigma+it)\log n}) = n^{-\sigma}\cos(t \log n)$. Hence, with $\theta_n = t \log n$,

$$\begin{aligned}\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| &= 3\operatorname{Re}[\log\zeta(\sigma)] + 4\operatorname{Re}[\log\zeta(\sigma + it)] + \operatorname{Re}[\log\zeta(\sigma + 2it)] \\ &= \sum c_n n^{-\sigma}(3 + 4\cos\theta_n + \cos 2\theta_n) \ge 0,\end{aligned}$$

by Lemma 1.4 and $c_n \ge 0$. ∎

Proof of Theorem 1.2. Suppose $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$. ζ is holomorphic at $1 + it_0$ and vanishes there to order ≥ 1, so $|\zeta(\sigma + it_0)|^4 \le C(\sigma - 1)^4$ as $\sigma \to 1$. Since $s = 1$ is a simple pole, $|\zeta(\sigma)|^3 \le C'(\sigma - 1)^{-3}$ as $\sigma \to 1$. And $|\zeta(\sigma + 2it_0)|$ stays bounded as $\sigma \to 1$. Hence $|\zeta^3(\sigma)\zeta^4(\sigma + it_0)\zeta(\sigma + 2it_0)| \to 0$ as $\sigma \to 1^+$, contradicting Corollary 1.5 (the logarithm of a number in $(0, 1)$ is negative). ∎

1.1 Estimates for $1/\zeta(s)$

The proof uses the logarithmic derivative ζ'/ζ near $\mathrm{Re}(s) = 1$, so we need growth estimates for ζ' (Proposition 2.7) and for $1/\zeta$. The following is a quantitative version of Theorem 1.2.

Proposition 1.6

For every $\epsilon > 0$, we have $1/|\zeta(s)| \le c_\epsilon |t|^\epsilon$ when $s = \sigma + it$, $\sigma \ge 1$, and $|t| \ge 1$.

Proof. From Corollary 1.5, $|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \ge 1$ for $\sigma \ge 1$. With the estimate for ζ in Proposition 2.7 and $|\zeta(\sigma)| \le c/(\sigma-1)$,

$$|\zeta^4(\sigma+it)| \ge c|\zeta^{-3}(\sigma)||t|^{-\epsilon} \ge c'(\sigma-1)^3|t|^{-\epsilon},$$

so

$$|\zeta(\sigma+it)| \ge c'(\sigma-1)^{3/4}|t|^{-\epsilon/4} \qquad (\sigma \ge 1,\ |t| \ge 1). \tag{3}$$

Two cases, with a constant A to be chosen. If $\sigma - 1 \ge A|t|^{-5\epsilon}$, (3) gives $|\zeta(\sigma+it)| \ge A'|t|^{-4\epsilon}$, and replacing 4ϵ by ϵ gives the estimate. If $\sigma - 1 < A|t|^{-5\epsilon}$, choose $\sigma' > \sigma$ with $\sigma' - 1 = A|t|^{-5\epsilon}$. Then $|\zeta(\sigma+it)| \ge |\zeta(\sigma'+it)| - |\zeta(\sigma'+it) - \zeta(\sigma+it)|$, and by the mean value theorem and the bound on ζ' (Proposition 2.7), $|\zeta(\sigma'+it) - \zeta(\sigma+it)| \le c''|\sigma' - \sigma||t|^\epsilon \le c''(\sigma'-1)|t|^\epsilon$. With (3) at σ',

$$|\zeta(\sigma+it)| \ge c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} - c''(\sigma'-1)|t|^\epsilon.$$

Choose $A = (c'/(2c''))^4$; since $\sigma' - 1 = A|t|^{-5\epsilon}$, this makes $c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} = 2c''(\sigma'-1)|t|^\epsilon$, hence $|\zeta(\sigma+it)| \ge c''(\sigma'-1)|t|^\epsilon = A''|t|^{-4\epsilon}$. Replace 4ϵ by ϵ. ∎

2 Reduction to the functions ψ and ψ_1

Tchebychev introduced a function easier to handle than $\pi(x)$ but carrying the same information:

$$\psi(x) = \sum_{p^m \le x} \log p,$$

summed over integers of the form $p^m \le x$ (p prime, $m \ge 1$). Equivalently, with $\Lambda(n) = \log p$ if $n = p^m$ for some prime p and $m \ge 1$, and $\Lambda(n) = 0$ otherwise,

$$\psi(x) = \sum_{1 \le n \le x} \Lambda(n), \qquad \psi(x) = \sum_{p \le x} \left[\frac{\log x}{\log p}\right] \log p,$$

$[u]$ the greatest integer $\le u$ (since $p^m \le x$ iff $m \le \log x/\log p$).

Proposition 2.1

If $\psi(x) \sim x$ as $x \to \infty$, then $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. We show $1 \le \liminf \pi(x)\frac{\log x}{x}$ and $\limsup \pi(x)\frac{\log x}{x} \le 1$ (4). Crudely, $\psi(x) = \sum_{p\le x}\left[\frac{\log x}{\log p}\right]\log p \le \sum_{p\le x}\frac{\log x}{\log p}\log p = \pi(x)\log x$, so $\frac{\psi(x)}{x} \le \frac{\pi(x)\log x}{x}$, and $\psi(x) \sim x$ gives the first inequality. For the second, fix $0 < \alpha < 1$:

$$\psi(x) \ge \sum_{p \le x} \log p \ge \sum_{x^\alpha < p \le x} \log p \ge (\pi(x) - \pi(x^\alpha))\alpha \log x,$$

so $\psi(x) + \alpha\pi(x^\alpha)\log x \ge \alpha\pi(x)\log x$. Dividing by x, and using $\pi(x^\alpha) \le x^\alpha$, $\alpha < 1$, $\psi(x) \sim x$, we get $1 \ge \alpha \limsup \pi(x)\frac{\log x}{x}$. As $\alpha < 1$ is arbitrary, the proof is complete. ∎

(The converse also holds.) It is more convenient to work with

$$\psi_1(x) = \int_1^x \psi(u)\,du.$$

Proposition 2.2

If $\psi_1(x) \sim x^2/2$ as $x \to \infty$, then $\psi(x) \sim x$ as $x \to \infty$, and therefore $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. By Proposition 2.1 it suffices to show $\psi(x) \sim x$. Since ψ is increasing, for $\alpha < 1 < \beta$,

$$\frac{1}{(1-\alpha)x}\int_{\alpha x}^{x} \psi(u)\,du \le \psi(x) \le \frac{1}{(\beta-1)x}\int_{x}^{\beta x} \psi(u)\,du.$$

So $\psi(x) \le \frac{1}{(\beta-1)x}[\psi_1(\beta x) - \psi_1(x)]$, i.e. $\frac{\psi(x)}{x} \le \frac{1}{\beta-1}\left[\frac{\psi_1(\beta x)}{(\beta x)^2}\beta^2 - \frac{\psi_1(x)}{x^2}\right]$, whence

$$\limsup_{x\to\infty} \frac{\psi(x)}{x} \le \frac{1}{\beta-1}\Big[\frac{1}{2}\beta^2 - \frac{1}{2}\Big] = \frac{1}{2}(\beta+1).$$

As $\beta > 1$ is arbitrary, $\limsup \psi(x)/x \le 1$; a similar argument with $\alpha < 1$ gives $\liminf \psi(x)/x \ge 1$. ∎

To relate ψ_1 and ζ: differentiating Lemma 1.3, for $\mathrm{Re}(s) > 1$, $\frac{\zeta'(s)}{\zeta(s)} = -\sum_{m,p}(\log p)p^{-ms} = -\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}$, i.e.

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}. \tag{5}$$

Proposition 2.3

For all $c > 1$,

$$\psi_1(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)\,ds. \tag{6}$$

Lemma 2.4

If $c > 0$, then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{a^s}{s(s+1)}\,ds = \begin{cases} 0 & \text{if } 0 < a \le 1,\\ 1 - 1/a & \text{if } 1 \le a.\end{cases}$$

Here, the integral is over the vertical line $\mathrm{Re}(s) = c$.

Proof. $|a^s| = a^c$ on the line, so the integral converges. Suppose $a \ge 1$, $a = e^\beta$, $\beta = \log a \ge 0$, and $f(s) = \frac{a^s}{s(s+1)} = \frac{e^{\beta s}}{s(s+1)}$; then $\mathrm{res}_{s=0} f = 1$ and $\mathrm{res}_{s=-1} f = -1/a$. For $T > 0$ let $\Gamma(T)$ be the vertical segment $S(T)$ from $c - iT$ to $c + iT$ together with the half-circle $C(T)$ of radius T centered at c, to the left of the segment, positively oriented (a toy contour). For T large, 0 and -1 are inside, and $\frac{1}{2\pi i}\int_{\Gamma(T)} f = 1 - 1/a$. On $C(T)$, for large T, $|s(s+1)| \ge \frac{1}{2}T^2$ and (as $\sigma \le c$) $|e^{\beta s}| \le e^{\beta c}$, so $\left|\int_{C(T)} f\right| \le \frac{C}{T^2}2\pi T \to 0$. If $0 < a \le 1$, use the half-circle to the right of $\mathrm{Re}(s) = c$; there are no poles inside and the same estimate applies (now $\beta \le 0$ and $\sigma \ge c$). ∎

Proof of Proposition 2.3. $\psi(u) = \sum_{n=1}^{\infty}\Lambda(n)f_n(u)$ with $f_n(u) = 1$ if $n \le u$ and 0 otherwise. Hence

$$\psi_1(x) = \int_0^x \psi(u)\,du = \sum_{n=1}^{\infty}\Lambda(n)\int_0^x f_n(u)\,du = \sum_{n\le x}\Lambda(n)(x-n).$$

By (5) and Lemma 2.4 with $a = x/n$,

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)ds = x\sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{(x/n)^s}{s(s+1)}\,ds = x\sum_{n\le x}\Lambda(n)\Big(1-\frac{n}{x}\Big) = \psi_1(x).$$

■

2.1 Proof of the asymptotics for ψ_1

We show $\psi_1(x) \sim x^2/2$, which completes the proof of the prime number theorem. The ingredients: the formula (6); the non-vanishing $\zeta(1+it) \neq 0$; and the estimates of Proposition 2.7 and Proposition 1.6. Strategy: move the line of integration from $\mathrm{Re}(s) = c$ to $\mathrm{Re}(s) = 1$, so that x^{s+1} has size x^2 instead of x^{c+1}; the pole of ζ at $s = 1$ contributes exactly the main term $x^2/2$, and the rest must be shown to be essentially smaller.

Fix $c = 2$, $x \ge 2$, and let

$$F(s) = \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big).$$

Deform the line $\mathrm{Re}(s) = c$ to the path $\gamma(T)$ ($T \ge 3$, to be chosen): along $\mathrm{Re}(s) = 1$ for $|t| \ge T$, and around $s = 1$ through the segment $\mathrm{Re}(s) = c$, $|t| \le T$, joined by horizontal segments at height $\pm T$. Then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} F(s)\,ds = \frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds. \tag{7}$$

Indeed, by Proposition 2.7 and Proposition 1.6, $|\zeta'(s)/\zeta(s)| \le A_\eta |t|^\eta$ for any fixed $\eta > 0$, when $\sigma \ge 1$, $|t| \ge 1$; so $|F(s)| \le A'|t|^{-2+\eta}$ in the two infinite regions between the line and $\gamma(T)$, where F is regular, and Cauchy's theorem gives (7).

Next pass from $\gamma(T)$ to $\gamma(T,\delta)$, which consists of γ_1: $\mathrm{Re}(s) = 1$, $t \le -T$; γ_2: horizontal segment at $t = -T$ from 1 to $1-\delta$; γ_3: $\mathrm{Re}(s) = 1-\delta$, $|t| \le T$; γ_4: horizontal segment at $t = T$ from $1-\delta$ to 1; γ_5: $\mathrm{Re}(s) = 1$, $t \ge T$. For fixed T, choose $\delta > 0$ so small that ζ has no zeros in the box $\{\sigma + it : 1-\delta \le \sigma \le 1,\ |t| \le T\}$ (possible since ζ does not vanish on $\sigma = 1$). F has a simple pole at $s = 1$: by Corollary 2.6, $\zeta(s) = \frac{1}{s-1} + H(s)$ with H regular near 1, so $-\zeta'(s)/\zeta(s) = \frac{1}{s-1} + h(s)$ with h holomorphic near 1, and $\mathrm{res}_{s=1} F = x^2/2$. Thus

$$\frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds = \frac{x^2}{2} + \frac{1}{2\pi i}\int_{\gamma(T,\delta)} F(s)\,ds.$$

Let $\epsilon > 0$ be given. γ_1, γ_5. For $\sigma = 1$, $|x^{1+s}| = x^2$, and by Proposition 1.6 (with Proposition 2.7) $|\zeta'/\zeta| \le A|t|^{1/2}$, so $\big|\int_{\gamma_5} F\big| \le Cx^2\int_T^\infty \frac{|t|^{1/2}}{t^2}\,dt$; the integral converges, so we can choose T with $\big|\int_{\gamma_1} F\big|, \big|\int_{\gamma_5} F\big| \le \epsilon x^2$. γ_3. With T fixed, $|x^{1+s}| = x^{2-\delta}$ on γ_3, so $\big|\int_{\gamma_3} F\big| \le C_T x^{2-\delta}$. γ_2, γ_4. $\big|\int_{\gamma_2} F\big| \le C'_T\int_{1-\delta}^1 x^{1+\sigma}\,d\sigma \le C'_T\frac{x^2}{\log x}$.

Hence there are constants C_T, C'_T with

$$\Big|\psi_1(x) - \frac{x^2}{2}\Big| \le \epsilon x^2 + C_T x^{2-\delta} + C'_T\frac{x^2}{\log x}, \qquad \Big|\frac{2\psi_1(x)}{x^2} - 1\Big| \le 2\epsilon + 2C_T x^{-\delta} + \frac{2C'_T}{\log x},$$

and for all large x, $\big|\frac{2\psi_1(x)}{x^2} - 1\big| \le 4\epsilon$. So $\psi_1(x) \sim x^2/2$, and the prime number theorem is proved.

Note on interchanging double sums

Proposition

Let $\{a_{k\ell}\}_{1\le k,\ell<\infty}$ be complex numbers with

$$\sum_{k=1}^{\infty}\Big(\sum_{\ell=1}^{\infty}|a_{k\ell}|\Big)<\infty. \tag{8}$$

Then:
(i) The double sum $A=\sum_{k=1}^{\infty}\sum_{\ell=1}^{\infty}a_{k\ell}$ converges, and the order of summation may be interchanged: $A=\sum_{\ell=1}^{\infty}\sum_{k=1}^{\infty}a_{k\ell}$.
(ii) Given $\epsilon>0$, there is N so that for all $K,L>N$, $|A-\sum_{k=1}^{K}\sum_{\ell=1}^{L}a_{k\ell}|<\epsilon$.
(iii) If $m\mapsto(k(m),\ell(m))$ is a bijection from $\mathbb{N}$ to $\mathbb{N}\times\mathbb{N}$ and $c_m=a_{k(m)\ell(m)}$, then $A=\sum_{m=1}^{\infty}c_m$.

Proof. (i) Let $b_k=\sum_\ell|a_{k\ell}|$, and let $S=\{x_0,x_1,\dots\}$ with $x_n\to x_0$. On S define $f_k(x_0)=\sum_{\ell=1}^{\infty}a_{k\ell}$, $f_k(x_n)=\sum_{\ell=1}^{n}a_{k\ell}$, and $g(x)=\sum_k f_k(x)$. Each f_k is continuous at x_0; $|f_k(x)|\le b_k$ with $\sum b_k<\infty$, so the series for g converges uniformly on S and g is continuous at x_0. Hence

$$\sum_k\sum_\ell a_{k\ell}=g(x_0)=\lim_n g(x_n)=\lim_n\sum_k\sum_{\ell=1}^{n}a_{k\ell}=\lim_n\sum_{\ell=1}^{n}\sum_k a_{k\ell}=\sum_\ell\sum_k a_{k\ell}.$$

(ii) $|A-\sum_{k\le K}\sum_{\ell\le L}a_{k\ell}|\le\sum_{k\le K}\sum_{\ell>L}|a_{k\ell}|+\sum_{k>K}\sum_\ell|a_{k\ell}|$. Choose N with $\sum_{k>N}b_k<\epsilon/2$; the first sum is $\le\sum_{k\le N}\sum_{\ell>L}|a_{k\ell}|+\sum_{k>N}b_k$, and for L large the finitely many tails $\sum_{\ell>L}|a_{k\ell}|$, $k\le N$, are small. [Supplement]

(iii) [Supplement]Given ϵ, take N as in (ii) (for the absolute values as well), and M so large that $\{c_1,\dots,c_M\}$ contains all $a_{k\ell}$ with $k,\ell\le N$. For $M'\ge M$, $\sum_{m\le M'}c_m-\sum_{k,\ell\le N}a_{k\ell}$ is a sum of terms $a_{k\ell}$ with $k>N$ or $\ell>N$, of total absolute value $<\epsilon$; so $|\sum_{m\le M'}c_m-A|<2\epsilon$. ∎